

Glycogen

Jason Ryan, MD, MPH

Glycogen

- Storage form of glucose
- Polysaccharide
- Repeating units of glucose
- Most abundant in muscle, liver
- Muscle: glycogen for own use
- Liver: glycogen for body

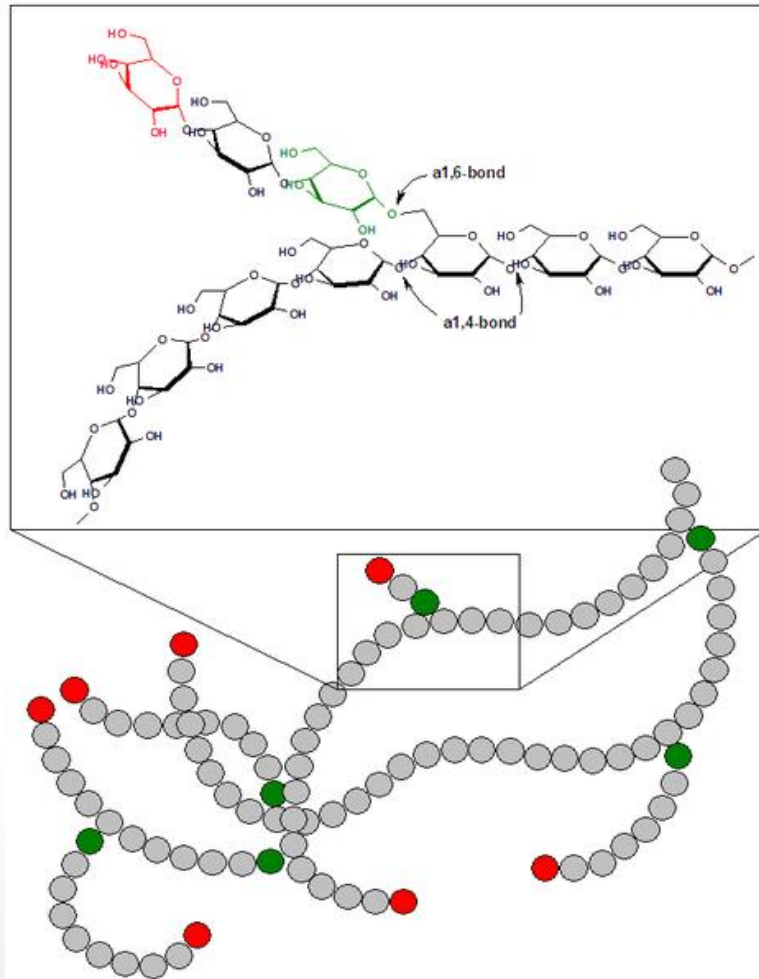


Lin Mei/Flickr

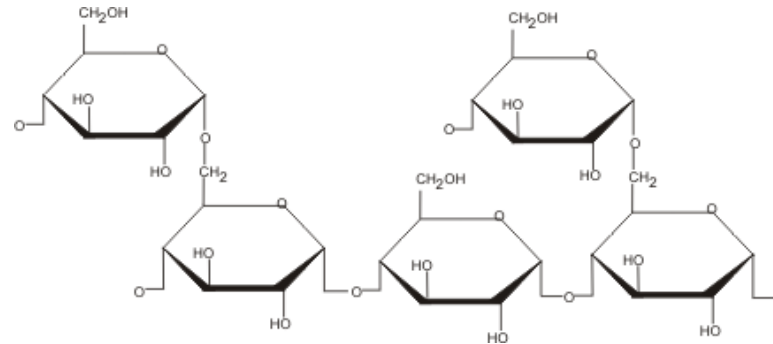


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Glycogen

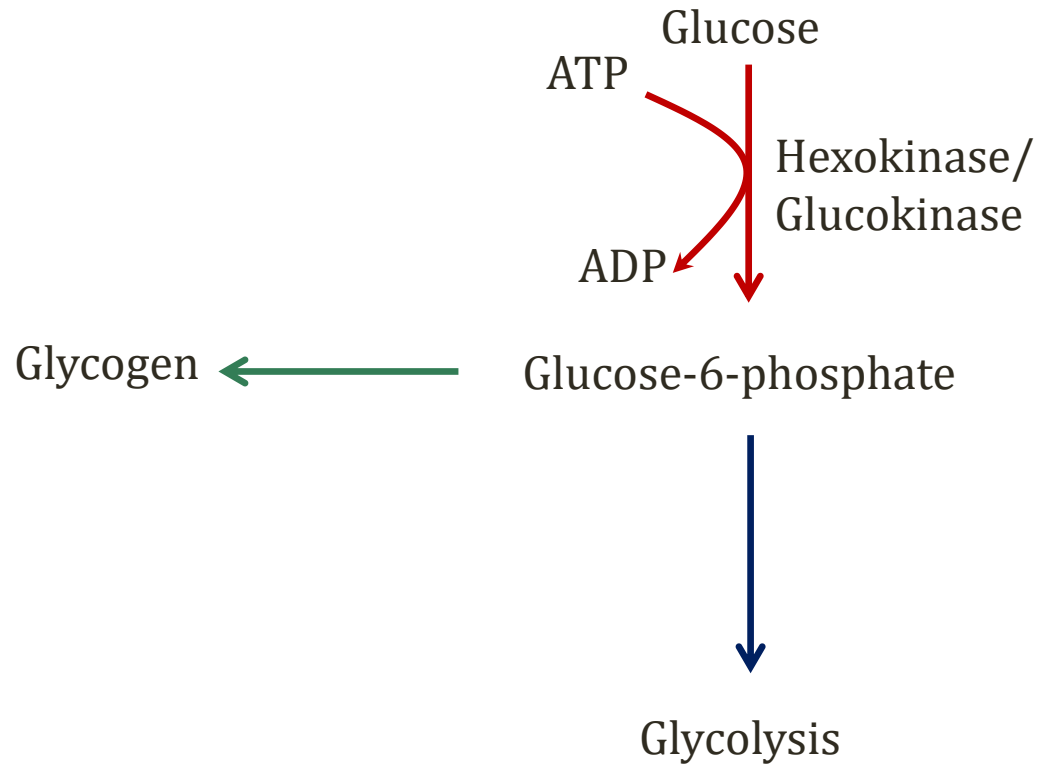


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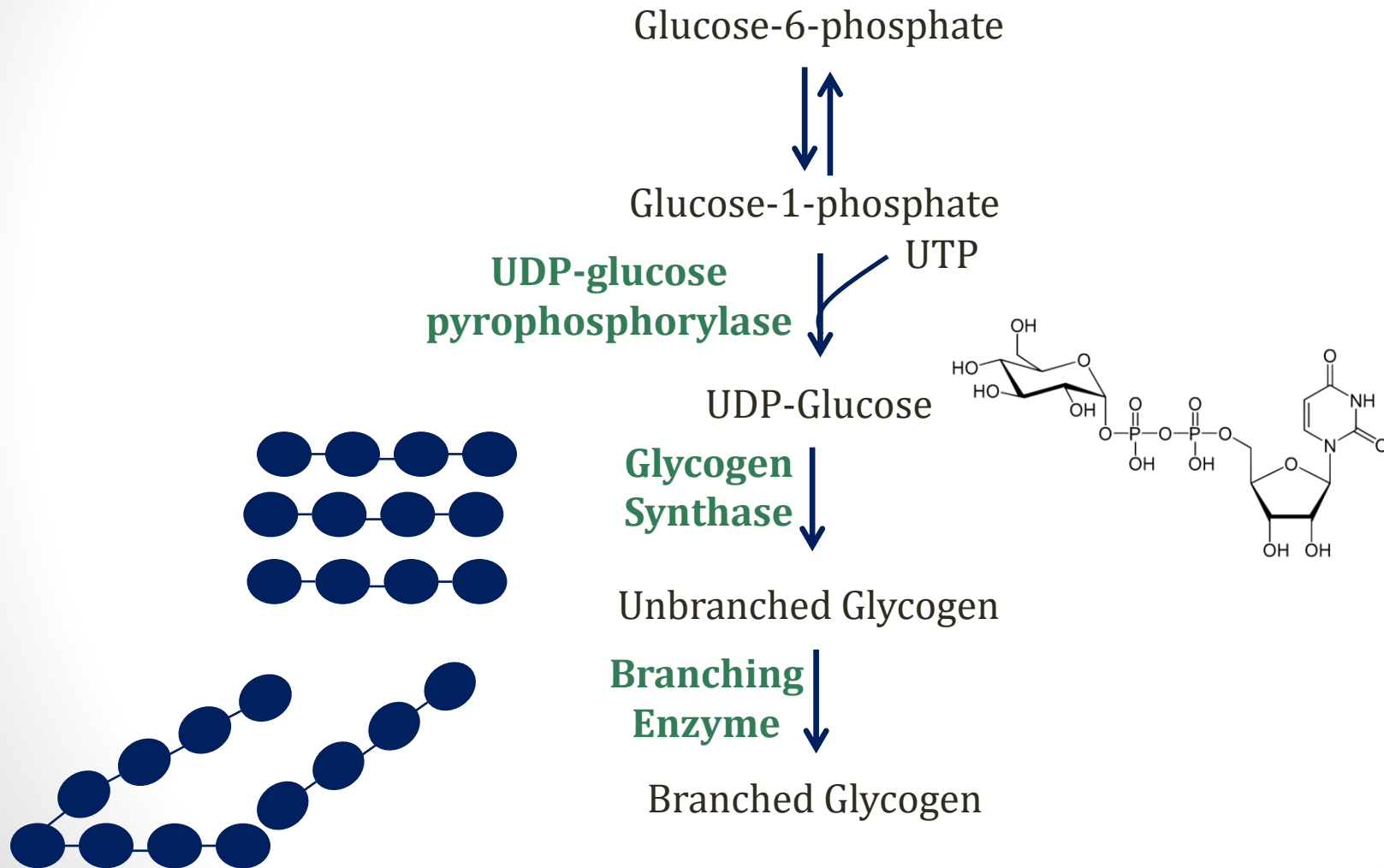


Boumpheyfr/Wikipedia

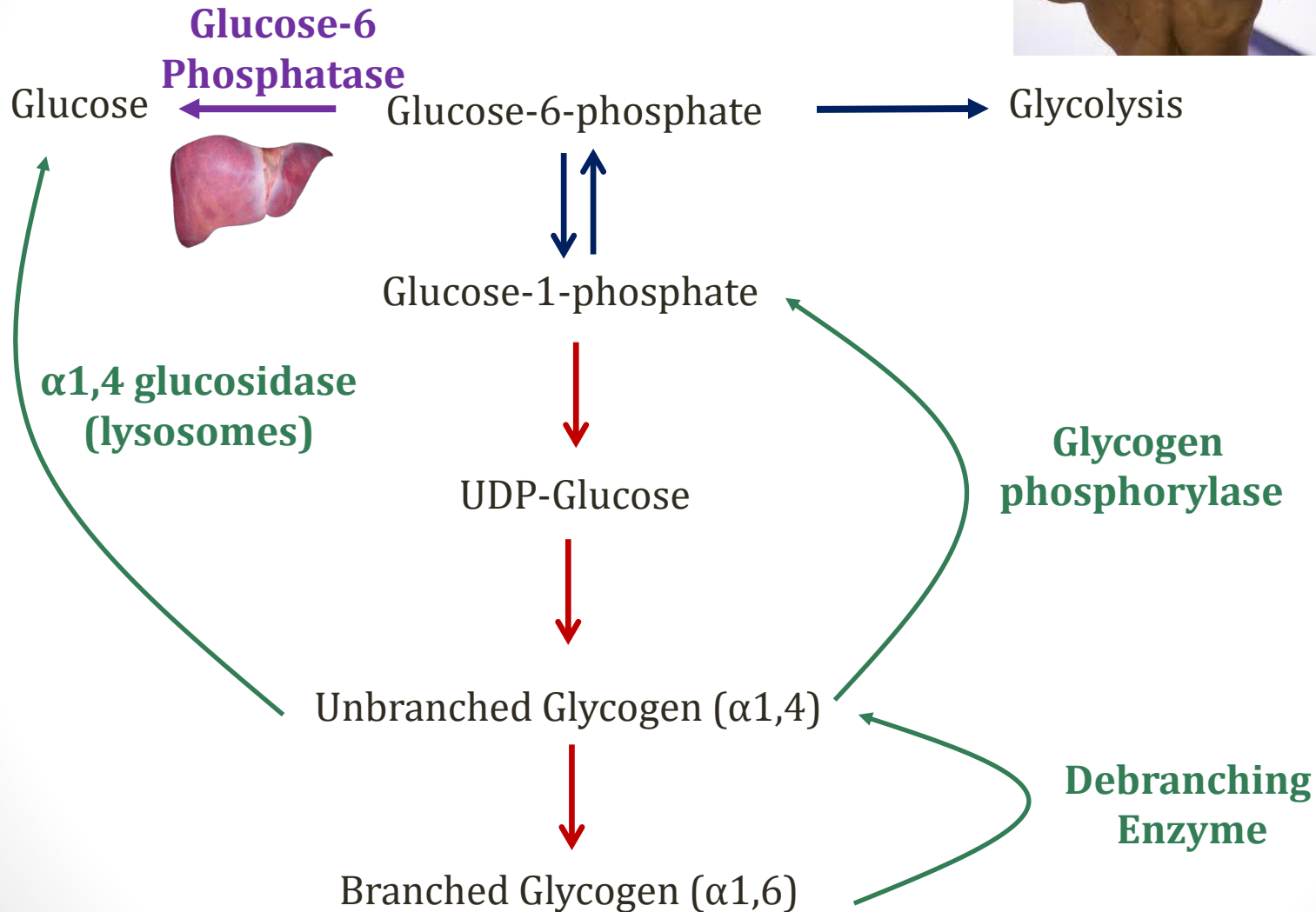
Glycogen Synthesis



Glycogen Synthesis



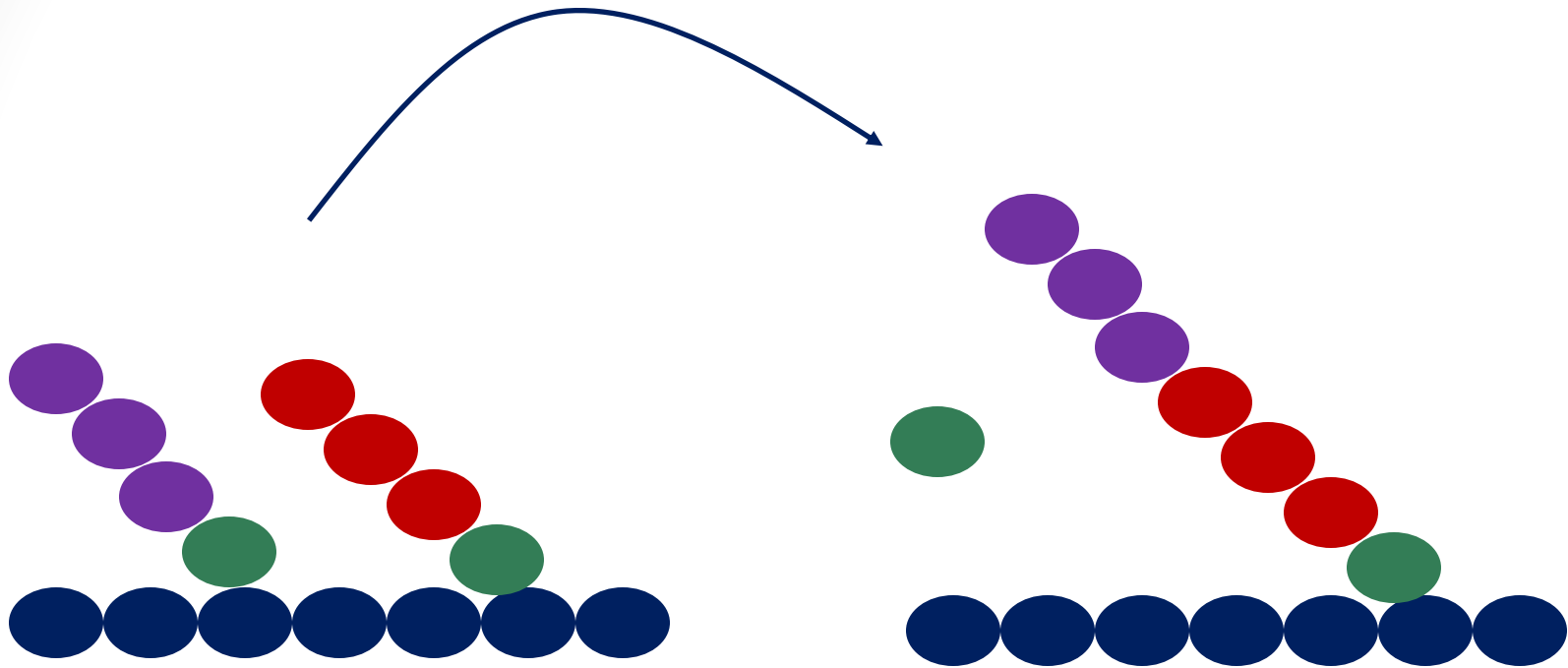
Glycogen Breakdown



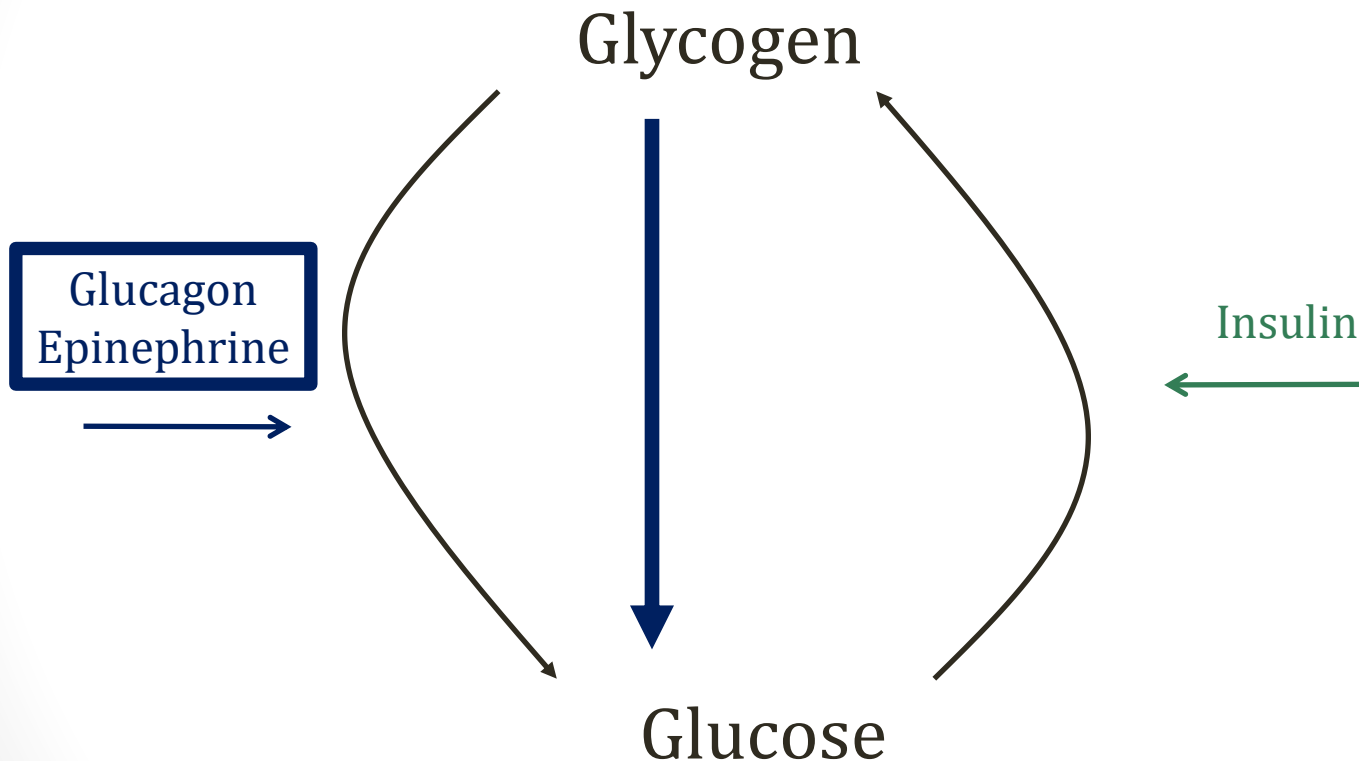
Glycogen Breakdown

- Phosphorylase
 - Removes glucose molecules from glycogen polymer
 - Creates glucose-1-phosphate
 - Stops when glycogen branches decreased to 2-4 linked glucose molecules (**limit dextrins**)
 - Stabilized by **vitamin B6**
- Debranching enzyme
 - Cleaves limit dextrins

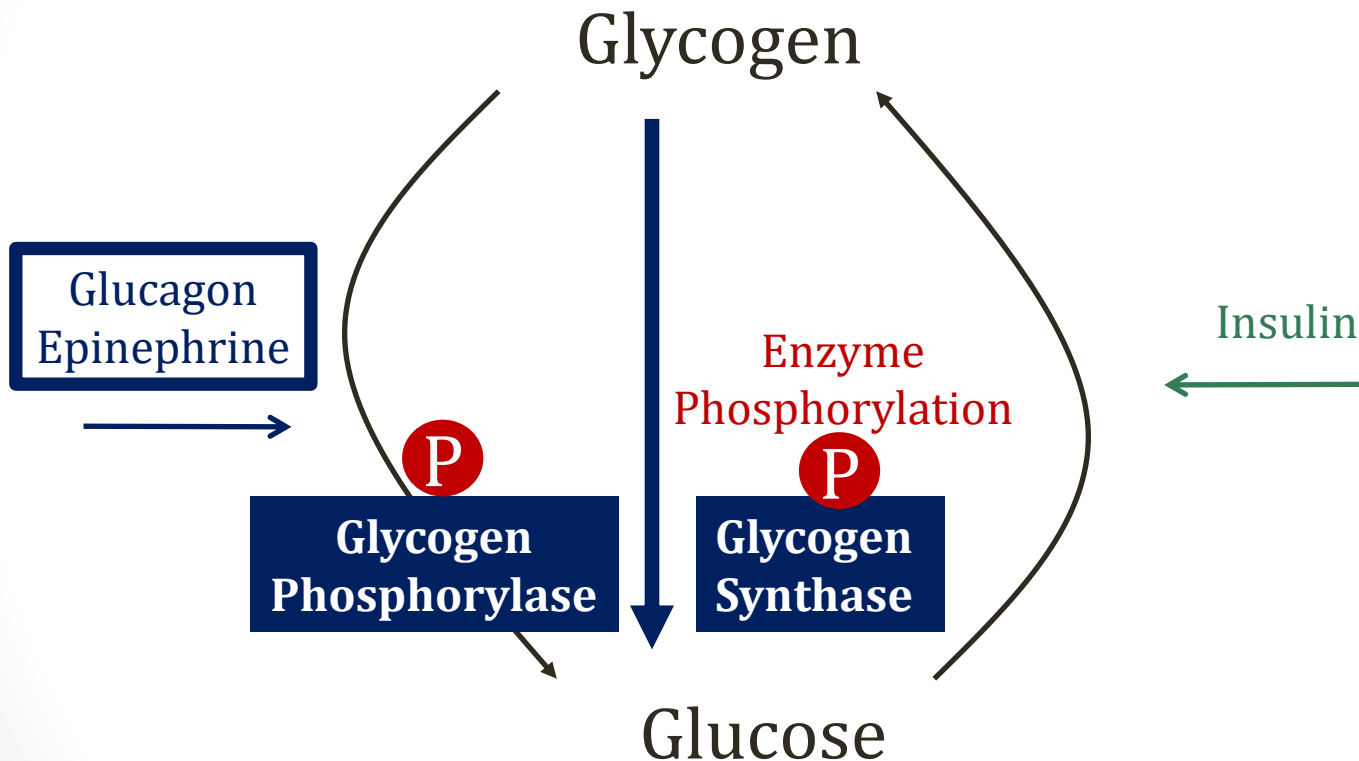
Debranching Enzyme



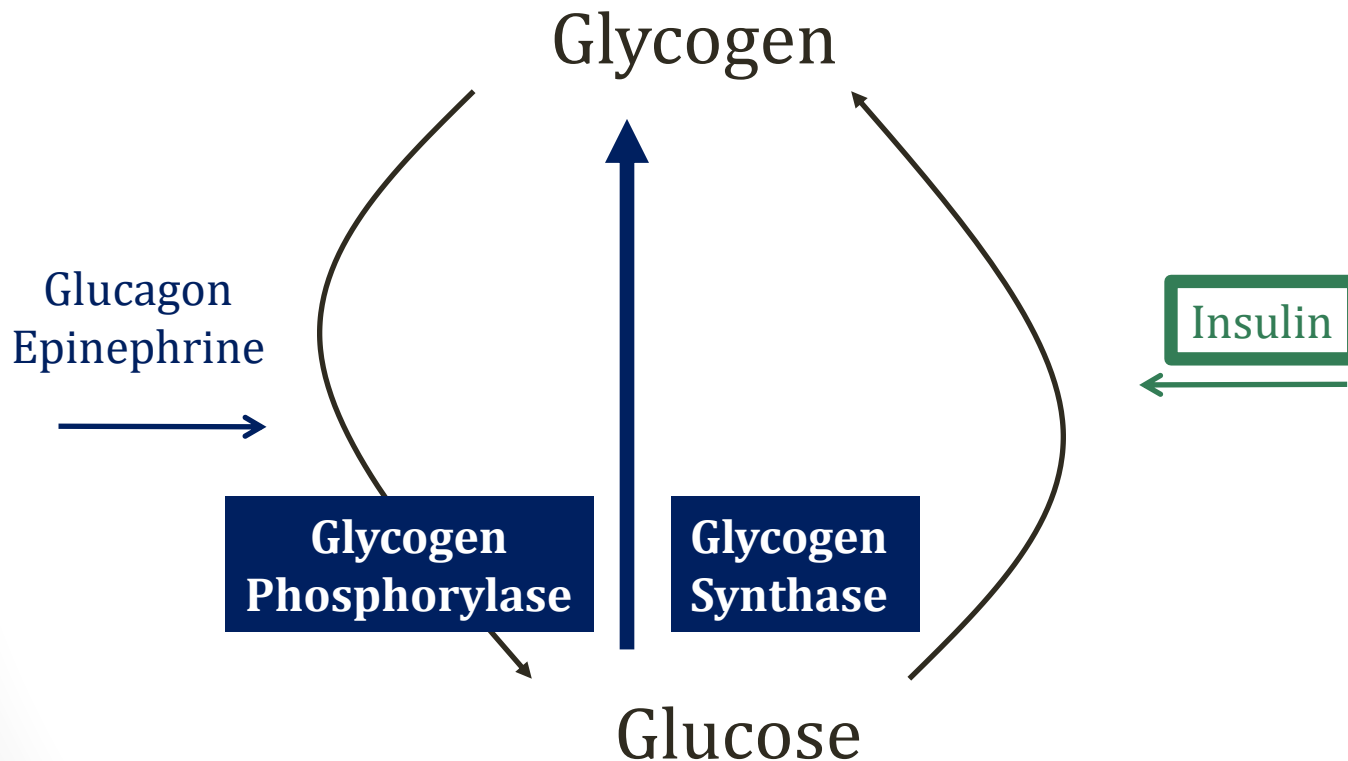
Hormonal Regulation



Hormonal Regulation

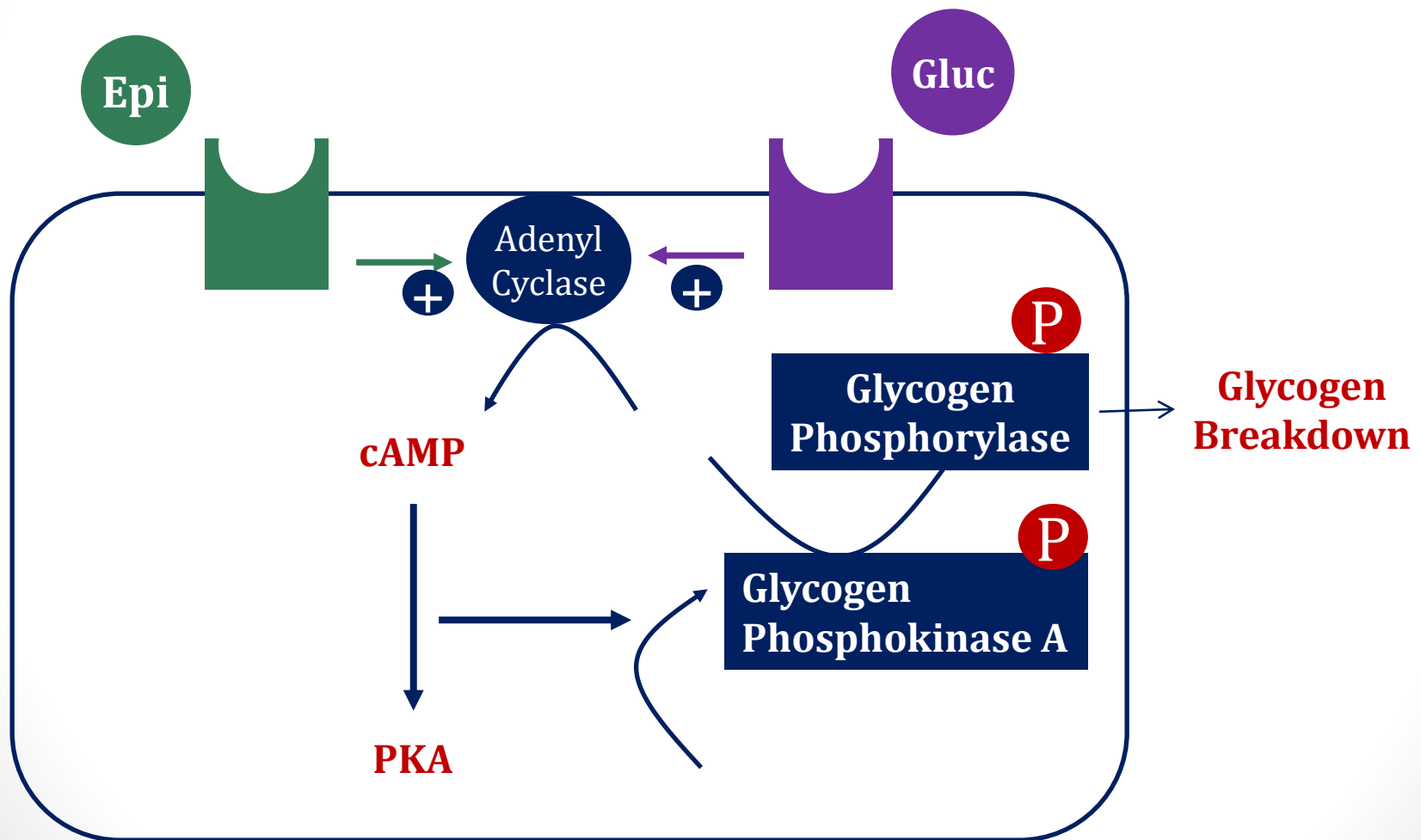


Hormonal Regulation



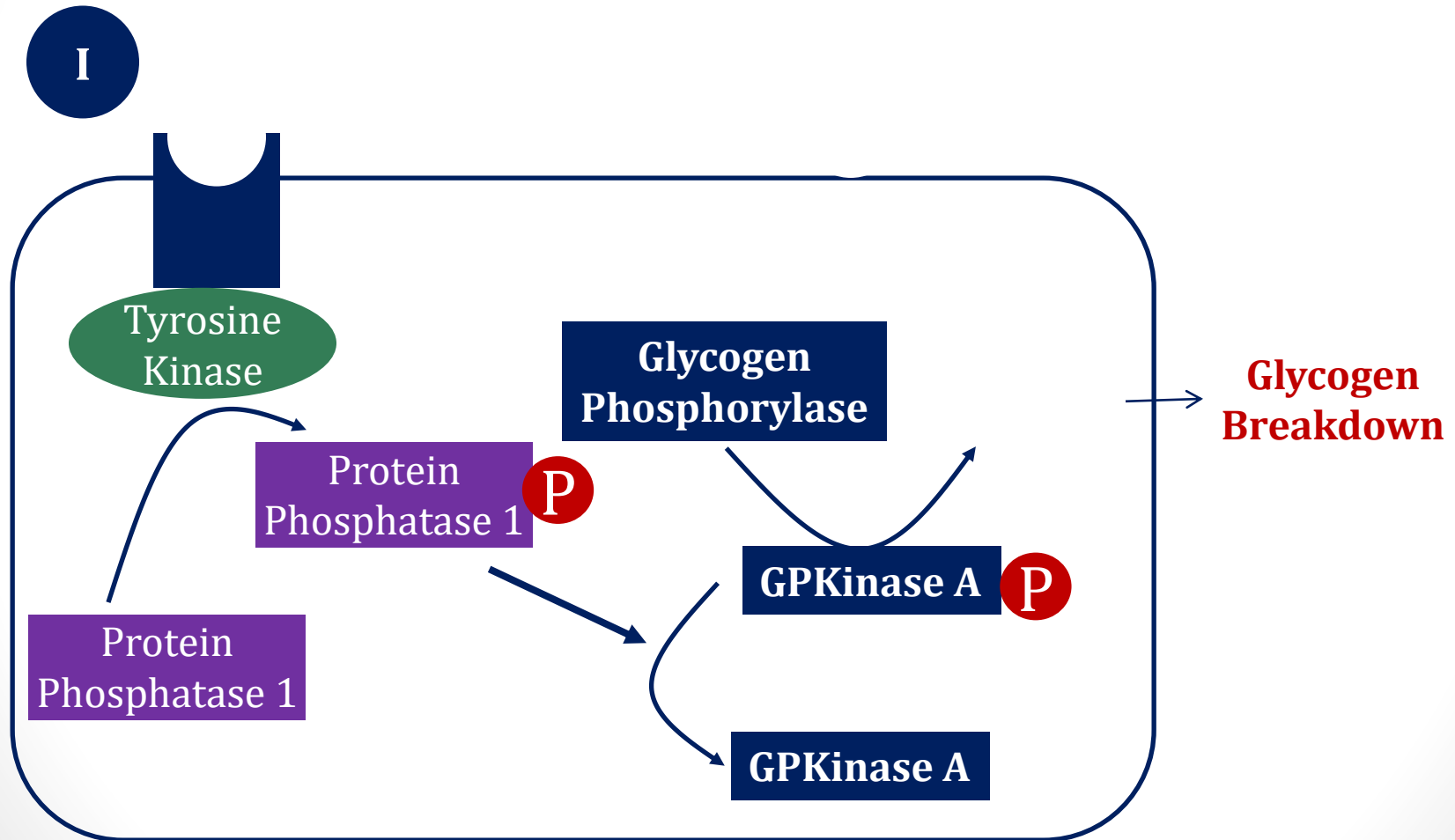
Epinephrine and Glucagon

Glycogen Phosphorylase



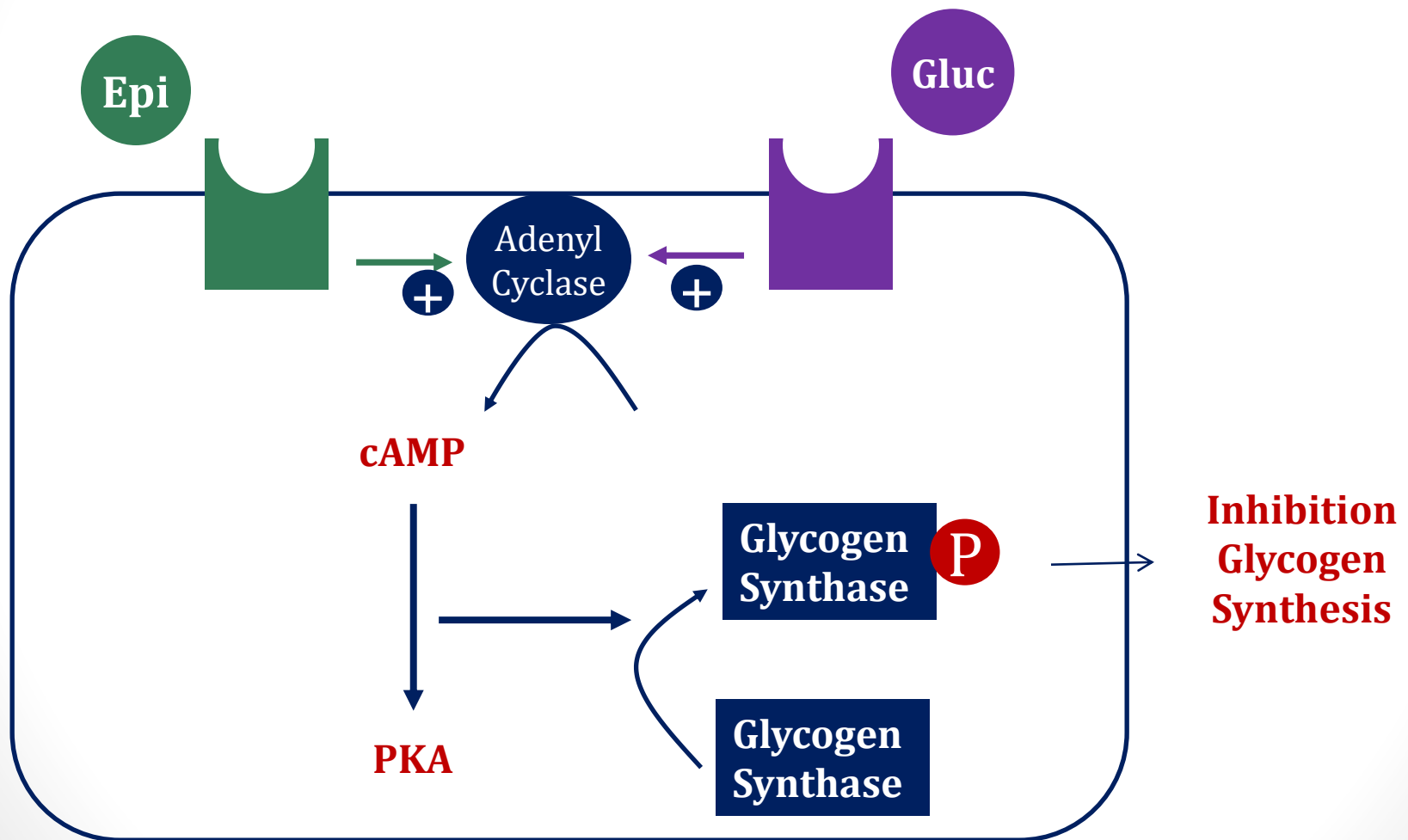
Insulin

Glycogen Phosphorylase



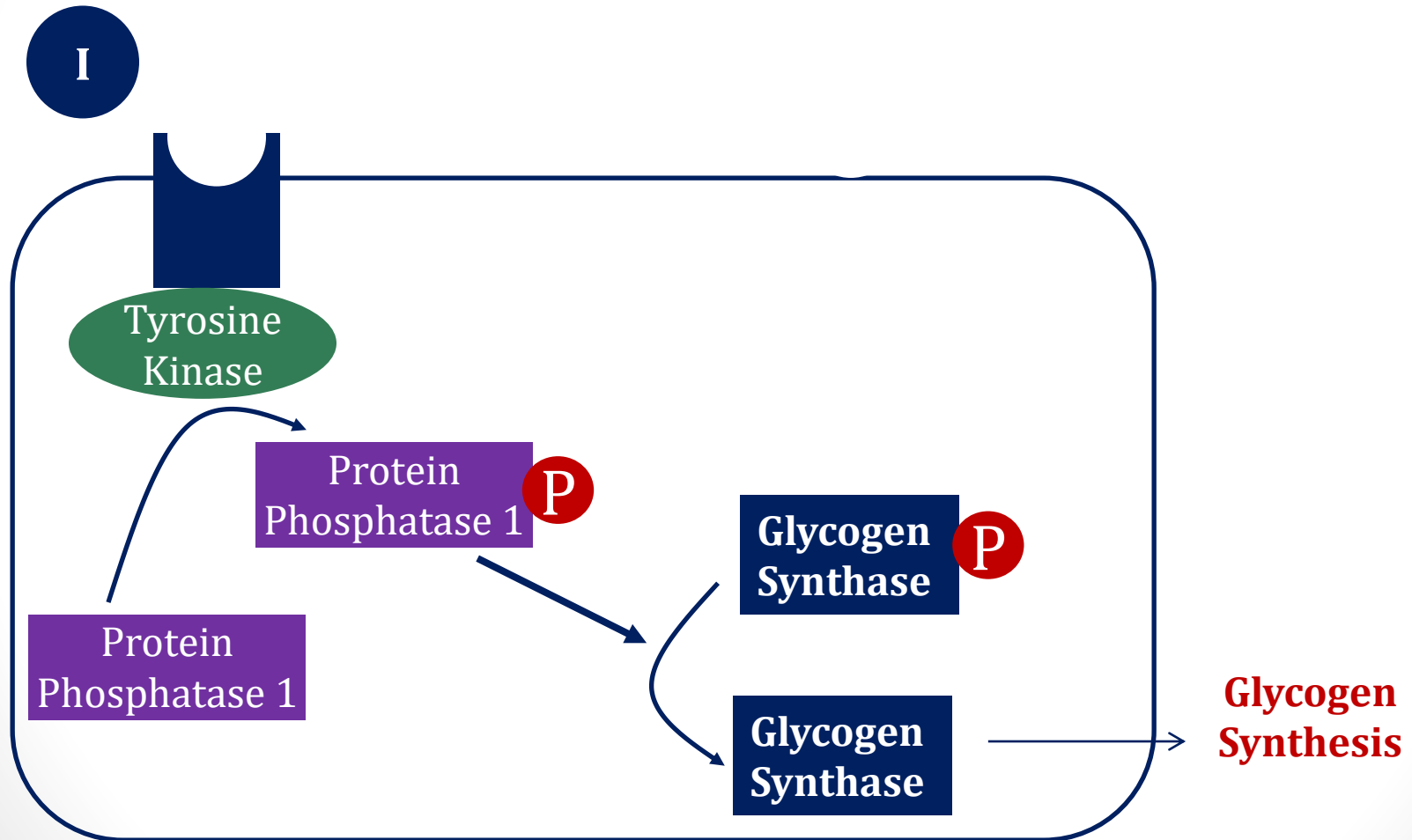
Epinephrine and Glucagon

Glycogen Synthase



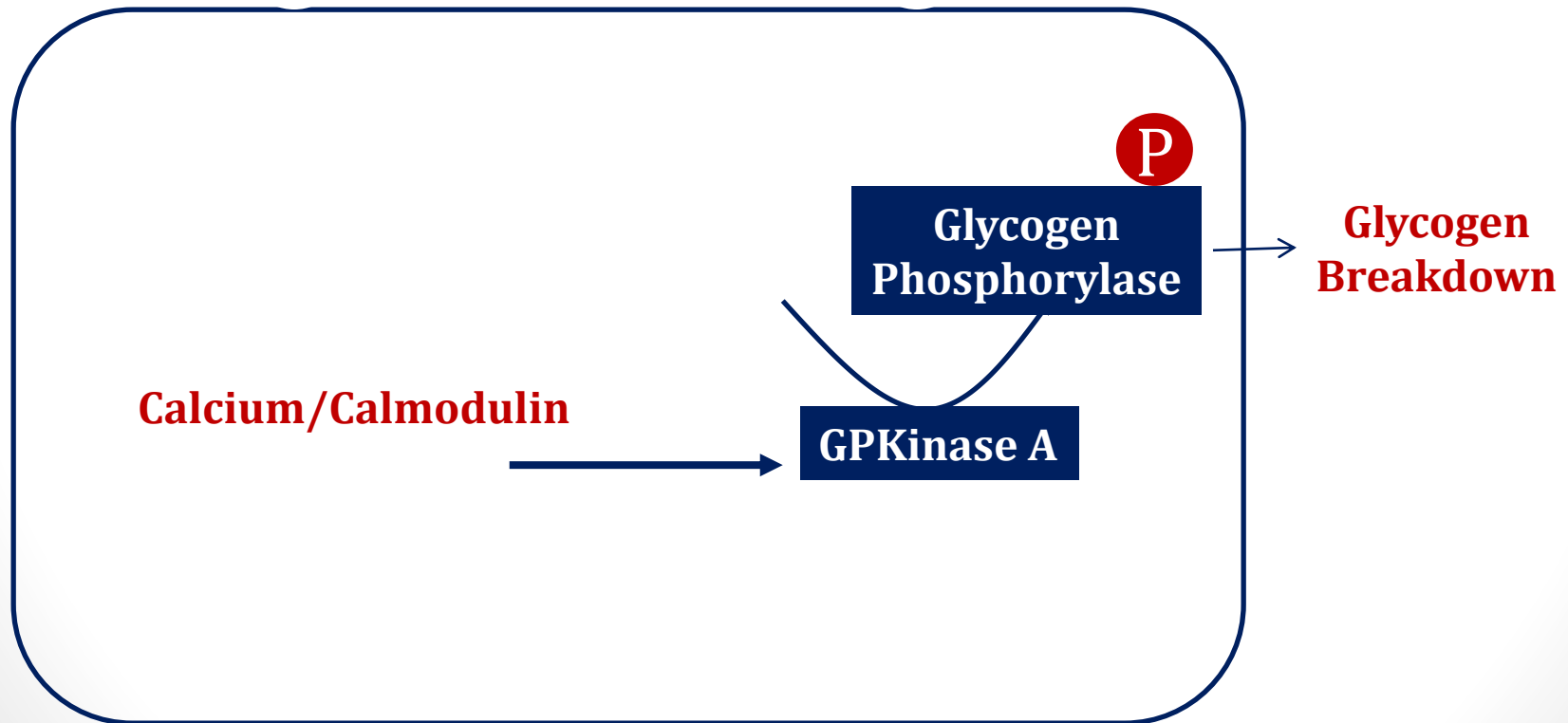
Insulin

Glycogen Synthase

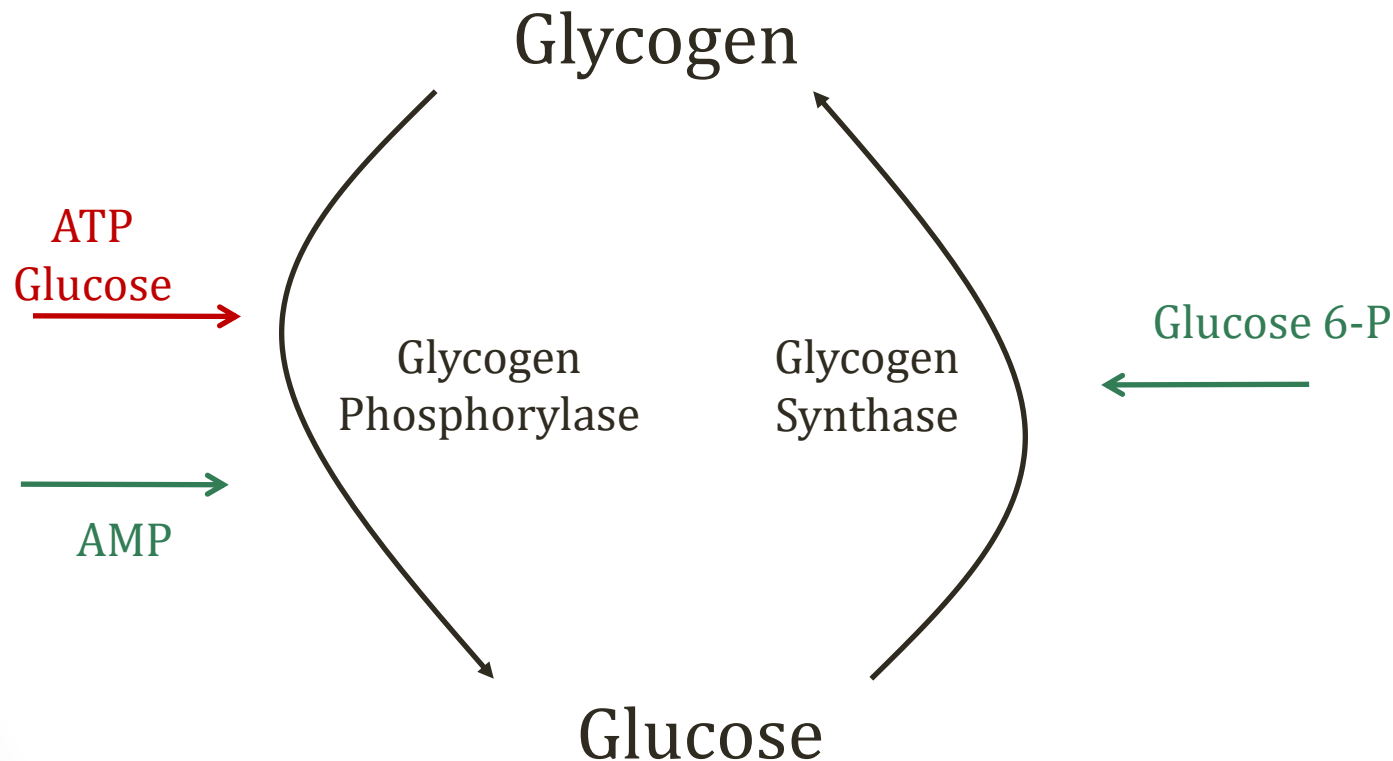


Muscle Contraction

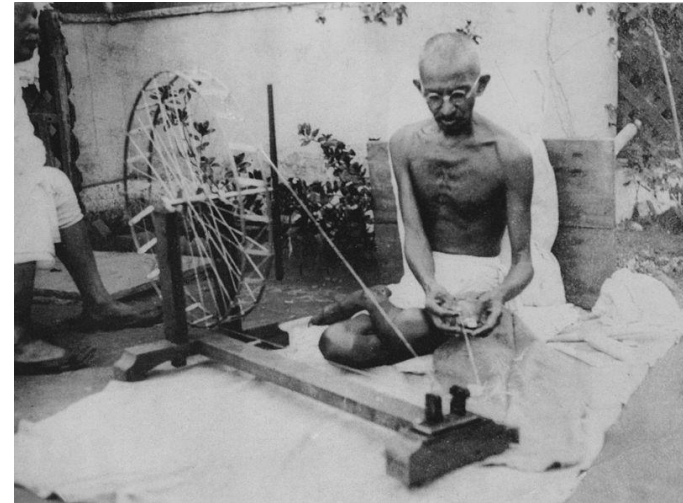
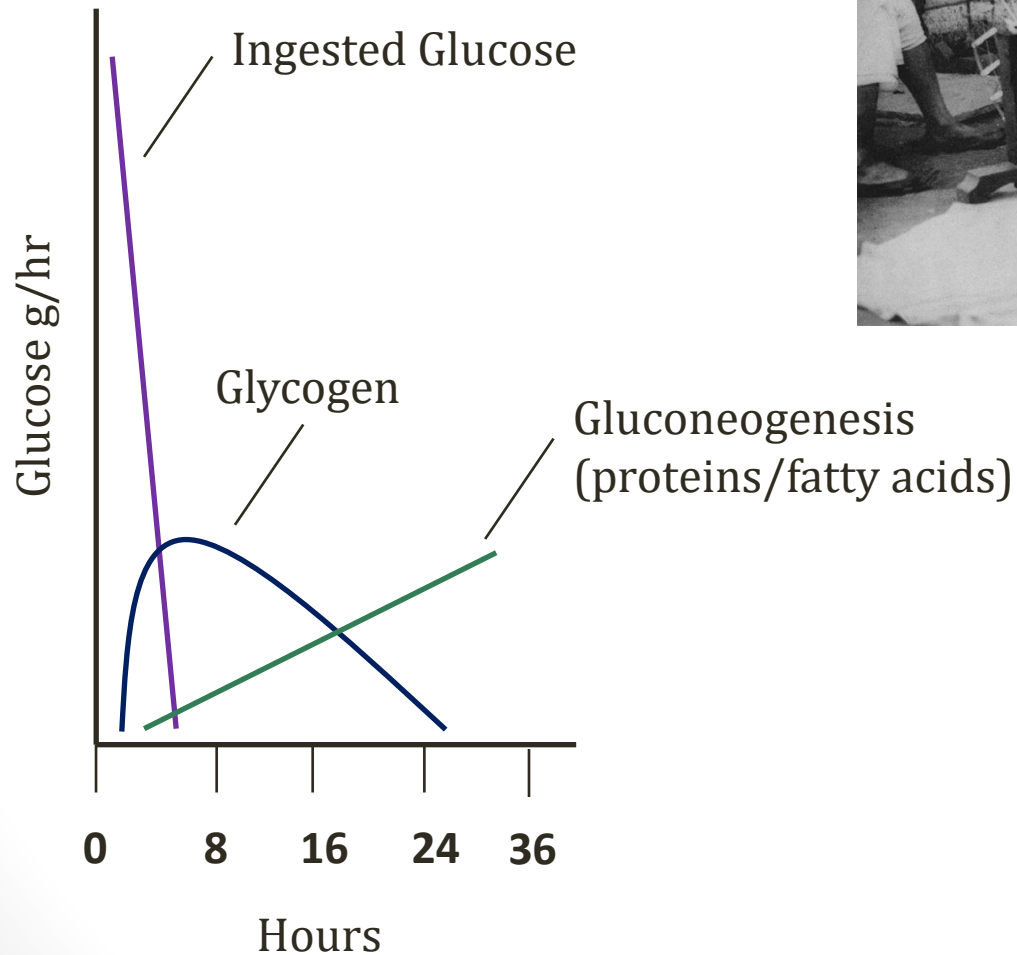
Glycogen Phosphorylase



Glycogen Regulation



Glycogen as Fuel



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Glycogen Storage Diseases

- Most autosomal recessive
- Defective breakdown of glycogen
- Liver: hypoglycemia
- Muscle: weakness
- More than 14 described

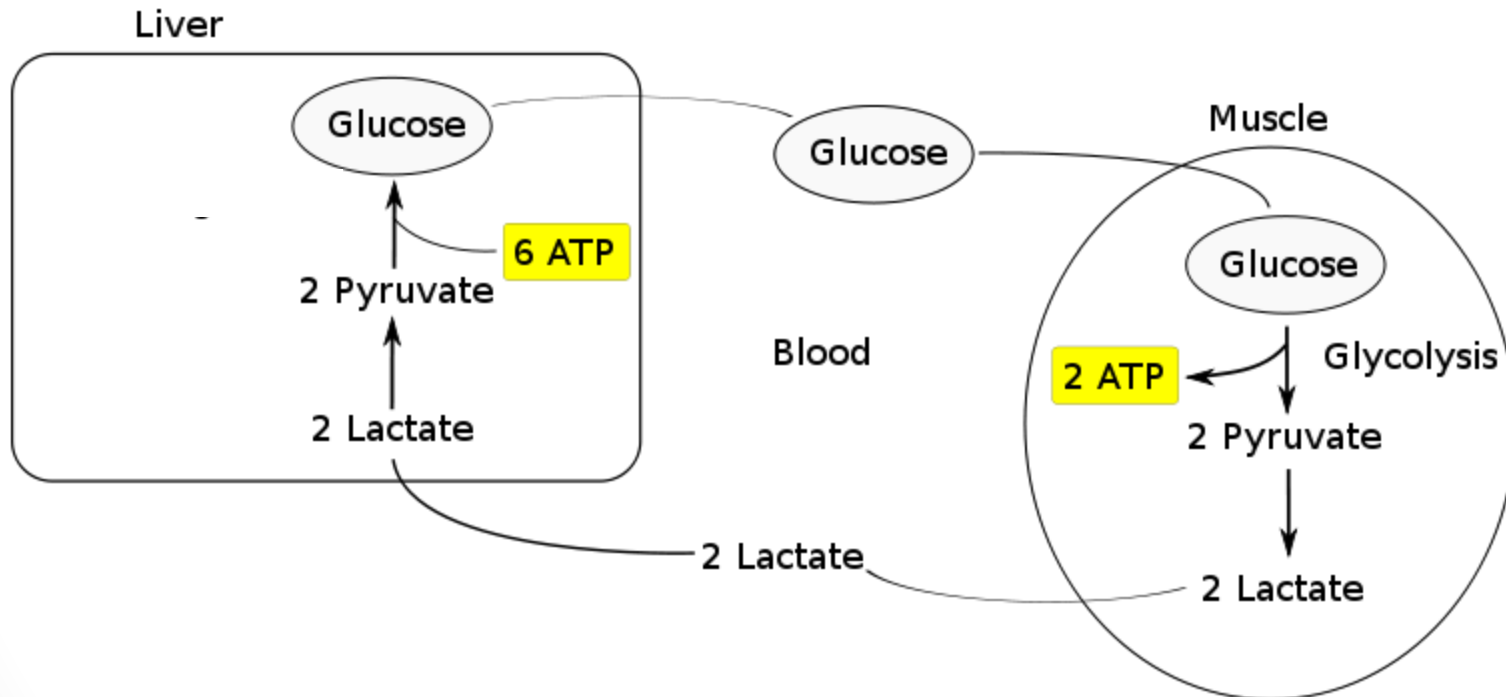
Von Gierke's Disease

Glycogen Storage Disease Type I

- Glucose-6-phosphatase deficiency (Type Ia)
 - Type Ib: Glucose transporter deficiency
- Presents in infancy: 2-6 months of age
- Severe hypoglycemia between meals
 - Lethargy
 - Seizures
 - Lactic acidosis (Cori cycle)
- Enlarged liver (excess glycogen)
 - Can lead to liver failure

Cori Cycle

Lactate Cycle



Petaholmes/Wikipedia

Von Gierke's Disease

Glycogen Storage Disease Type I

- Diagnosis:
 - DNA testing (preferred)
 - Liver biopsy (historical test)
- Treatment: Cornstarch (glucose polymer)
- Avoid sucrose, lactose, fructose, galactose
 - Feed into glycolysis pathways
 - Cannot be metabolized to glucose via gluconeogenesis
 - Worsen accumulation of glucose 6-phosphate

Pompe's Disease

Glycogen Storage Disease Type II

- Acid alpha-glucosidase deficiency
 - Also “lysosomal acid maltase”
- Accumulation of glycogen in lysosomes
- Classic form presents in infancy
- Severe disease → often death in infancy/childhood

Pompe's Disease

Glycogen Storage Disease Type II

- Enlarged muscles
 - Cardiomegaly
 - Enlarged tongue
- Hypotonia
- Liver enlargement (often from heart failure)
- No metabolic problems (hypoglycemia)
- Death from heart failure

Cori's Disease

Glycogen Storage Disease Type III

- Debranching enzyme deficiency
- Similar to type I except:
 - Milder hypoglycemia
 - No lactic acidosis (Cori cycle intact)
 - Muscle involvement (glycogen accumulation)
- Key point: Gluconeogenesis is **intact**

Cori's Disease

Glycogen Storage Disease Type III

- Classic presentation:
 - Infant or child with hypoglycemia/hepatomegaly
 - Hypotonia/weakness
 - Possible cardiomyopathy with hypertrophy

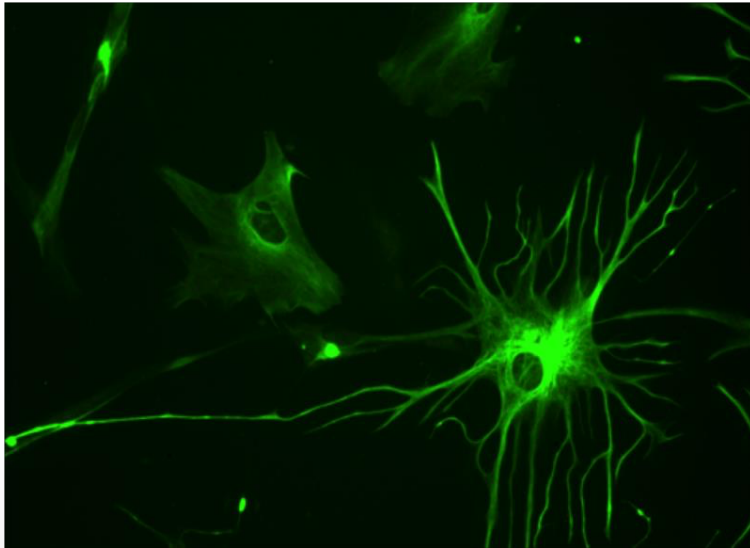
McArdle's Disease

Glycogen Storage Disease Type V

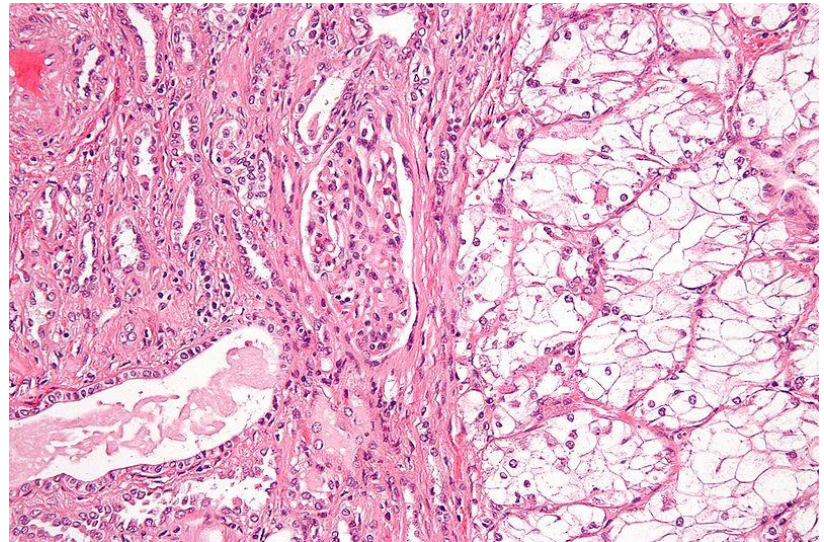
- Muscle glycogen phosphorylase deficiency
 - Myophosphorylase deficiency
 - Skeletal muscle has unique isoform of G-phosphorylase
- Glycogen not properly broken down in muscle cells
- Usually presents in **adolescence/early adulthood**
 - Exercise intolerance, fatigue, **cramps**
 - Poor endurance, muscle swelling, and weakness
 - **Myoglobinuria** and CK release (especially with exercise)
 - Urine may turn dark after exercise

Other Glycogen Locations

- Astrocytes
- Renal cell carcinoma



Bruno Pascal/Wikipedia



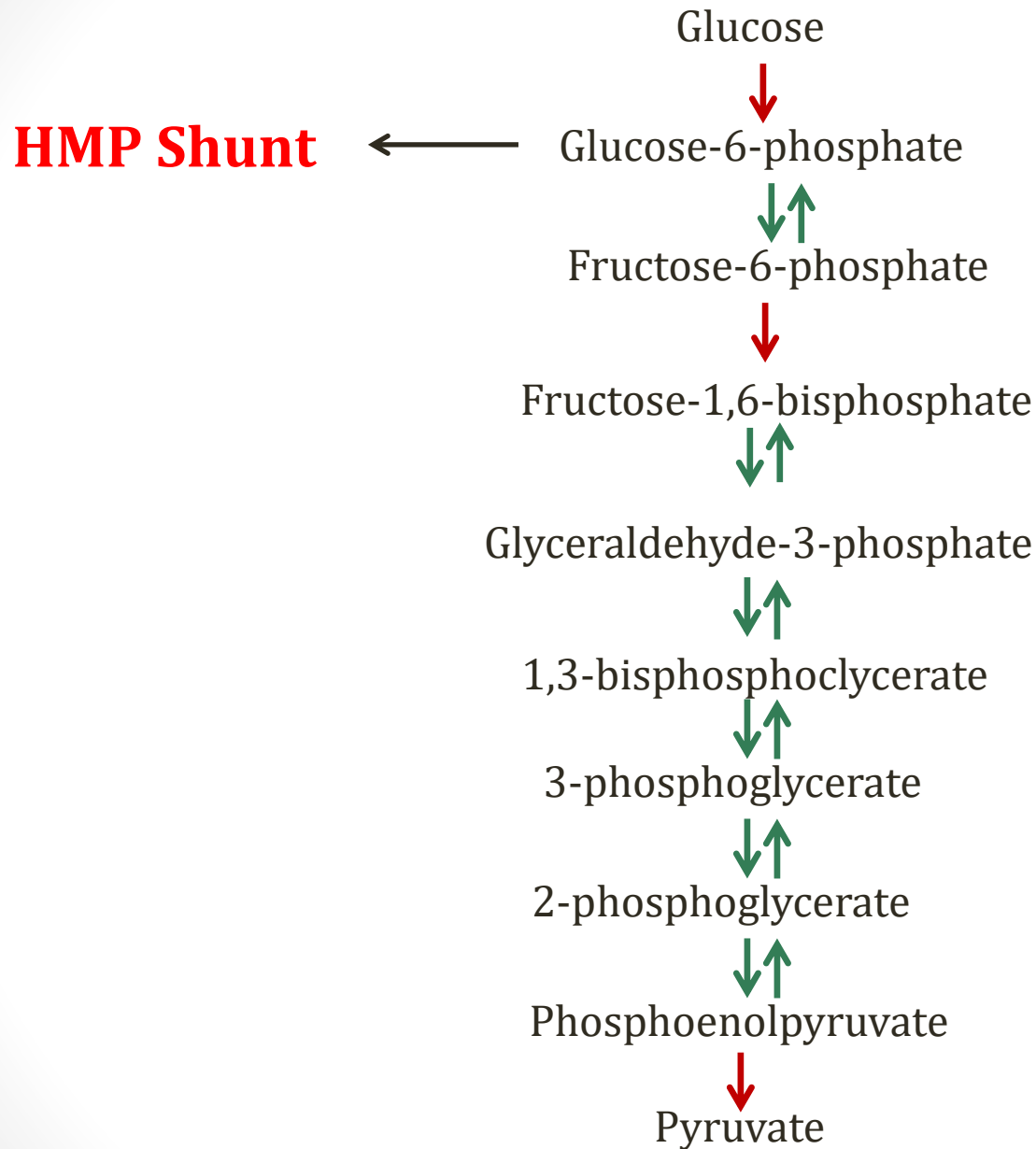
Nephron/Wikipedia

HMP Shunt

Jason Ryan, MD, MPH

HMP Shunt

- Series of reactions that goes by several names:
 - Hexose monophosphate shunt
 - Pentose phosphate pathway
 - 6-phosphogluconate pathway
- Glucose 6-phosphate “shunted” away from glycolysis

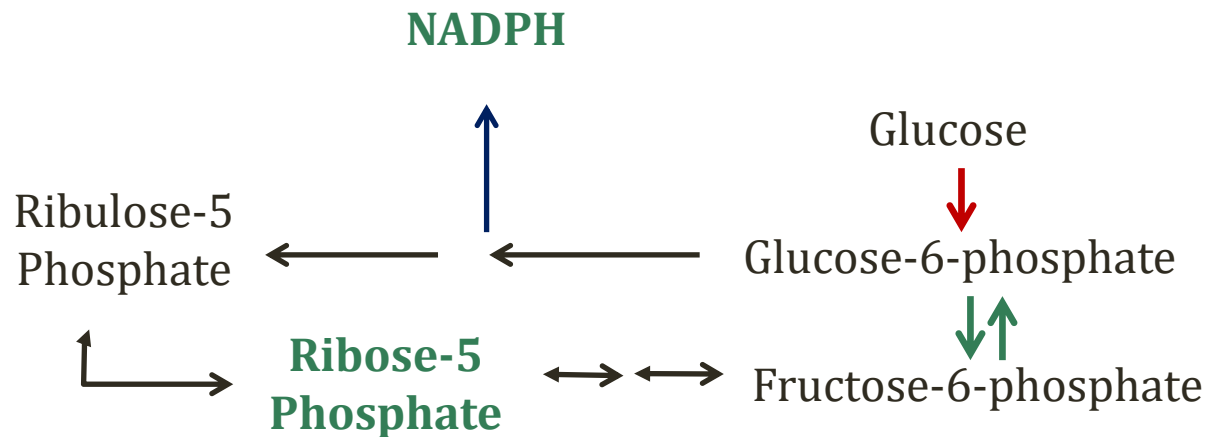


HMP Shunt

- Synthesizes:
 - **NADPH** (many uses)
 - **Ribose 5-phosphate** (nucleotide synthesis)
- Two key clinical correlations:
 - **G6PD deficiency**
 - **Thiamine deficiency (transketolase)**

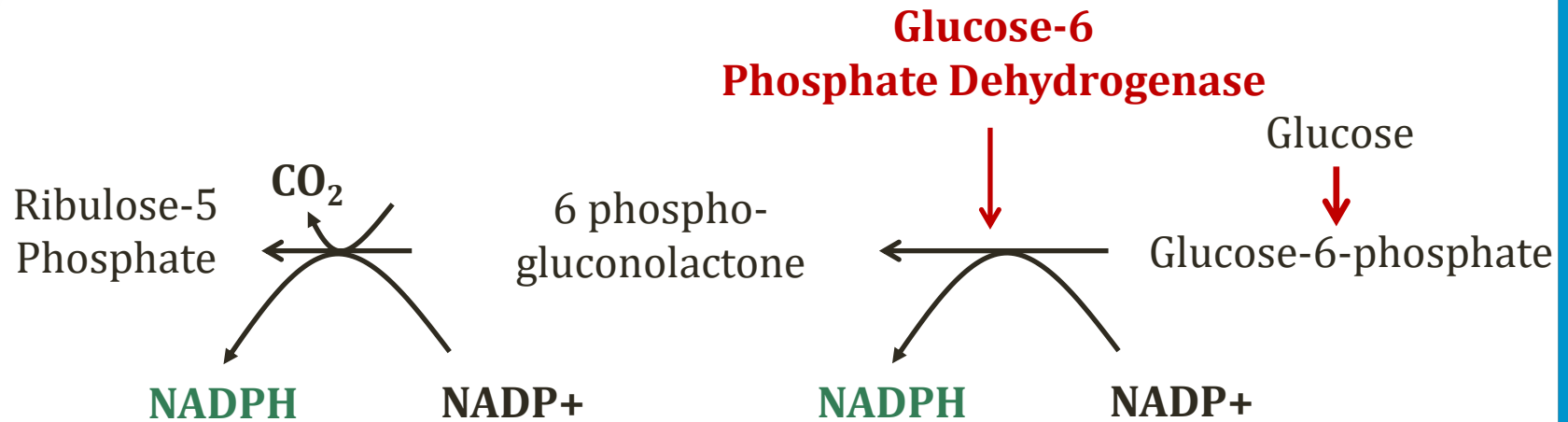
HMP Shunt

- All reactions occur in cytosol
- Two phases:
 - Oxidative: irreversible, rate-limiting
 - Reductive: reversible



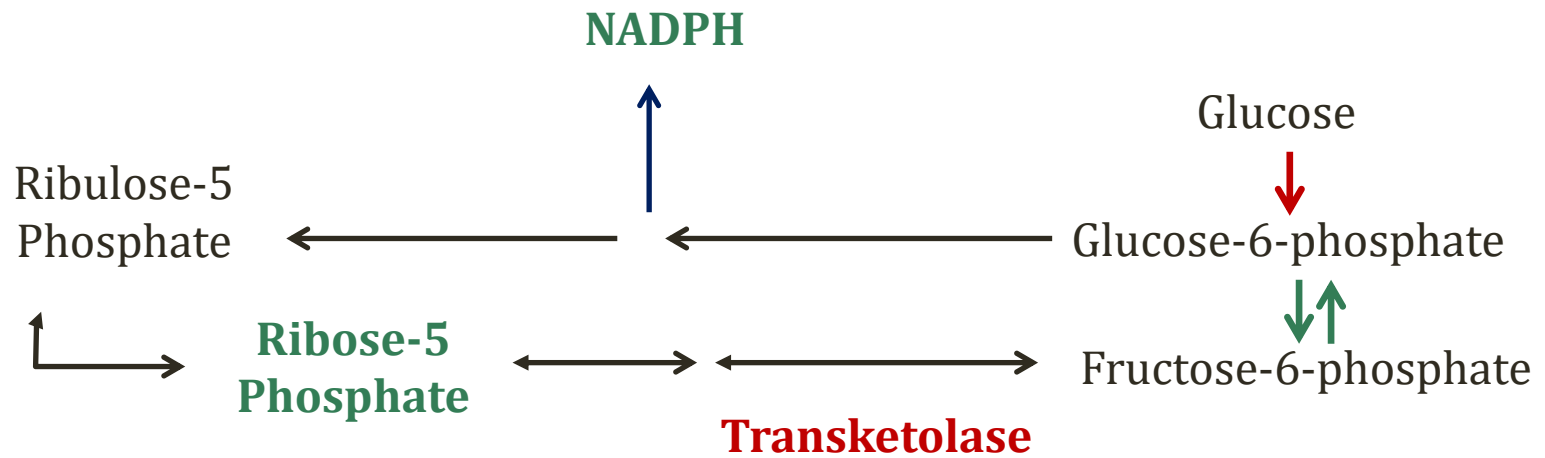
HMP Shunt

Oxidative Reactions



HMP Shunt

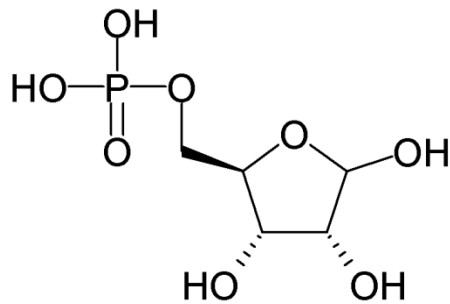
Reductive Reactions



Transketolase

- Transfers a carbon unit to create F-6-phosphate
- Requires **thiamine (B1)** as a co-factor
- **Wernicke-Korsakoff syndrome**
 - Abnormal transketolase may predispose
 - Affected individuals may have abnormal binding to thiamine

Ribose-5-Phosphate



Ribose 5-phosphate

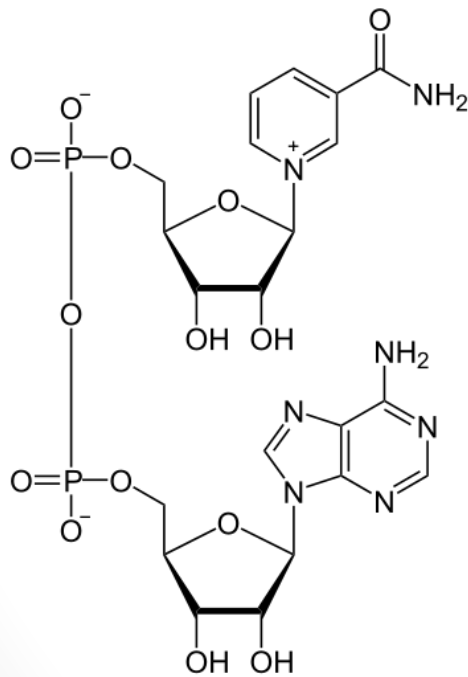
Purine Nucleotides
Adenosine, Guanosine

Pyrimidine Nucleotides
Cytosine, Uridine, Thymidine

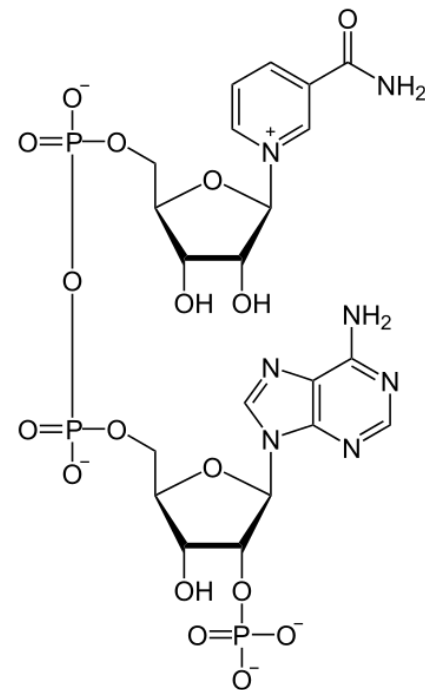
NADPH

Nicotinamide adenine dinucleotide phosphate

- Similar structure to NADH
- Not used for oxidative phosphorylation (ATP)



NADH



NADPH

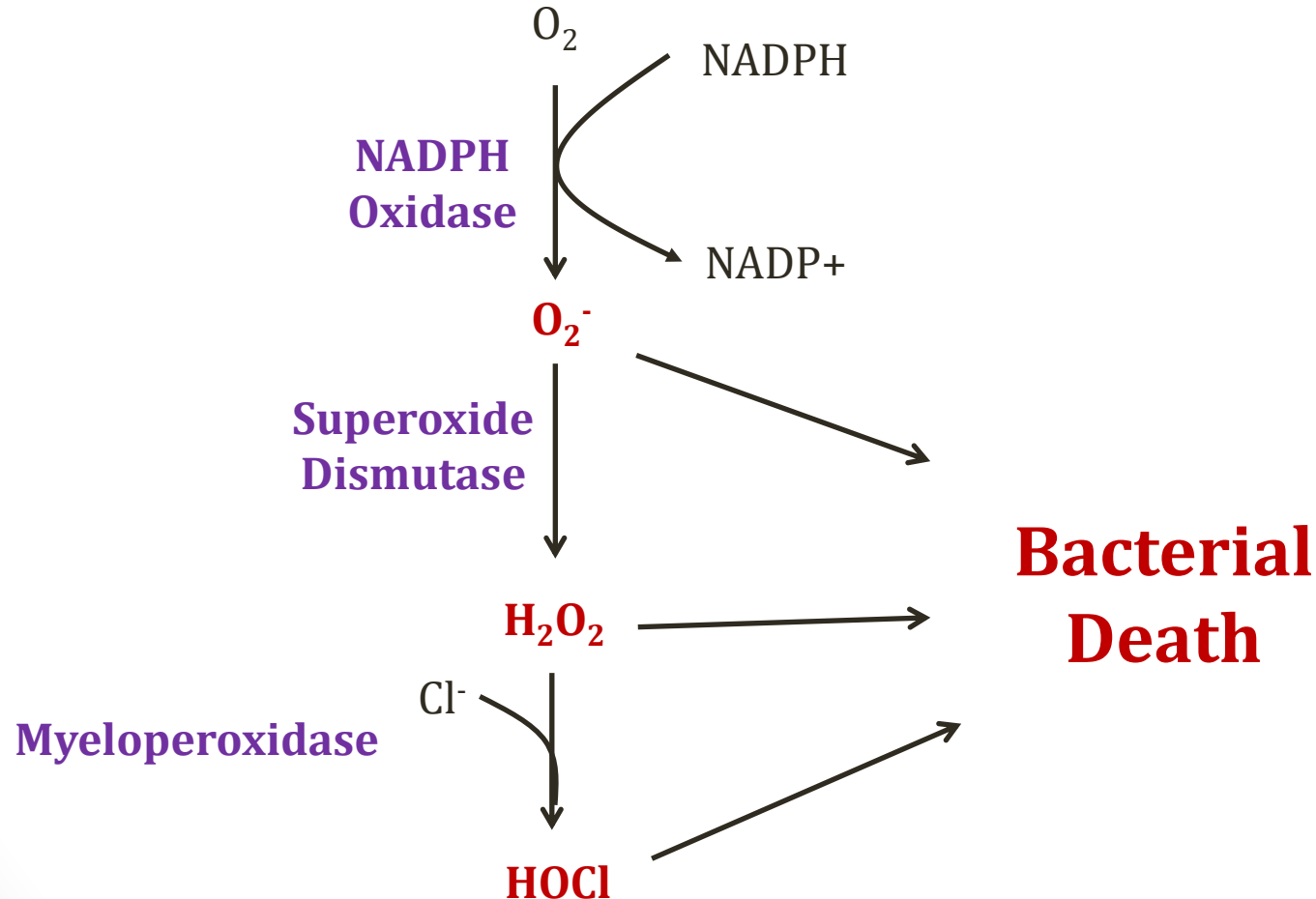
NADPH Uses

- Used in “reductive” reactions
- Releases hydrogen to form NADP^+
- Use #1: Co-factor in **fatty acid, steroid synthesis**
 - Liver, mammary glands, testis, adrenal cortex
- Use #2: **Phagocytosis**
- Use #3: Protection from **oxidative damage**

Respiratory Burst

- Phagocytes generate H_2O_2 to kill bacteria
 - “Oxygen dependent” killing
 - “Oxygen independent”: low pH, enzymes
- Uses three key enzymes:
 - NADPH oxidase
 - Superoxide dismutase
 - Myeloperoxidase

Respiratory Burst



CGD

Chronic Granulomatous Disease

- Loss of function of NADPH oxidase
- Phagocytes cannot generate H_2O_2
- Catalase (-) bacteria generate their own H_2O_2
 - Phagocytes use despite enzyme deficiency
- **Catalase (+) bacteria** breakdown H_2O_2
 - Host cells have no H_2O_2 to use → recurrent infections
- Five organisms cause almost all CGD infections:
 - Staph aureus, Pseudomonas, Serratia, Nocardia, Aspergillus

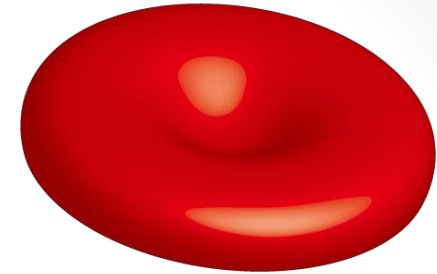
G6PD Deficiency

Glucose-6-Phosphate Dehydrogenase

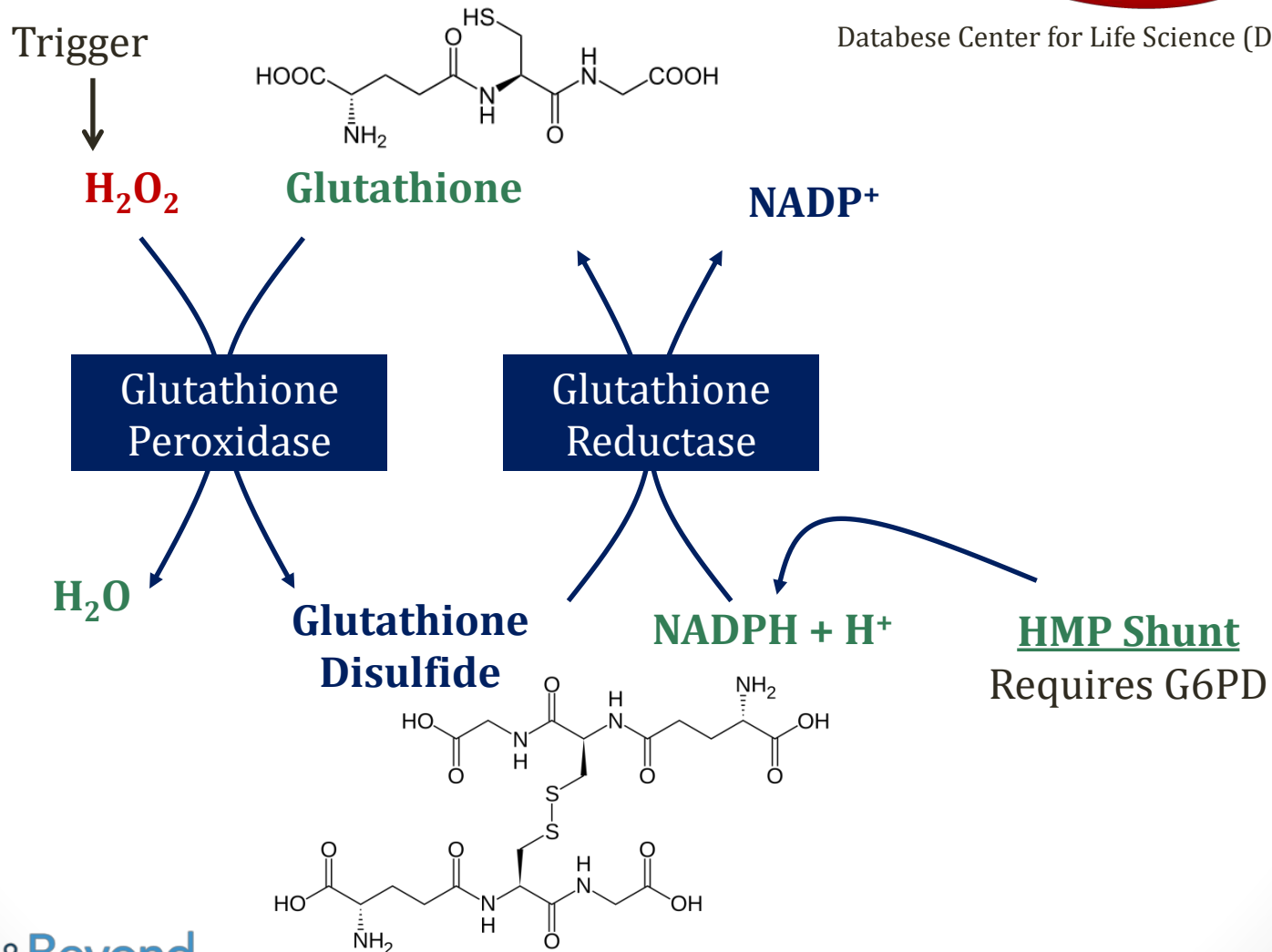
- NADPH required for normal red blood cell function
- H_2O_2 generation triggered in RBCs
 - Infections
 - Drugs
 - Fava beans
- Need NADPH to degrade H_2O_2
- Absence of required NADPH → hemolysis

Glutathione

Erythrocytes



Database Center for Life Science (DBCLS)



G6PD Deficiency

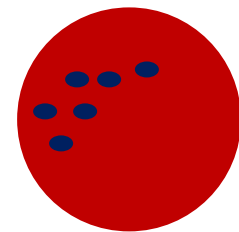
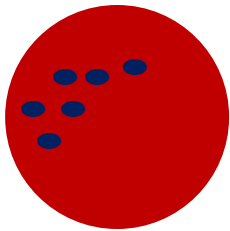
Glucose-6-Phosphate Dehydrogenase

- **X-linked** disorder (males)
- Most common human enzyme disorder
- High prevalence in **Africa**, Asia, the Mediterranean
 - May protect against malaria
- **Recurrent hemolysis** after exposure to trigger
May present as dark urine
- Other HMP functions usually okay
 - Nucleic acids, fatty acids, etc.

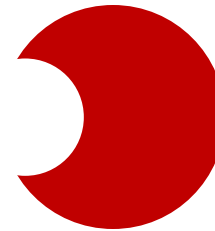
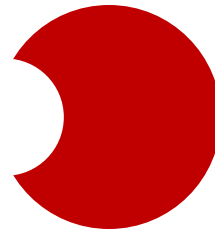
G6PD Deficiency

Glucose-6-Phosphate Dehydrogenase

- Classic findings: **Heinz bodies** and **bite cells**
- Heinz bodies: oxidized Hgb precipitated in RBCs
- Bite cells: phagocytic removal by splenic macrophages



Heinz bodies



Bite cells

G6PD Deficiency

Triggers

- Infection: Macrophages generate free radicals
- Fava beans: Contain oxidants
- Drugs:
 - Antibiotics (**sulfa drugs**, dapsone, nitrofurantoin, INH)
 - **Anti-malarials** (primaquine, quinidine)
 - Aspirin, acetaminophen (rare)

G6PD Deficiency

Diagnosis and Treatment

- Diagnosis:
 - Fluorescent spot test
 - Detects generation of NADPH from NADP
 - Positive test if blood spot fails to fluoresce under UV light
- Treatment:
 - Avoidance of triggers

Fructose and Galactose

Jason Ryan, MD, MPH

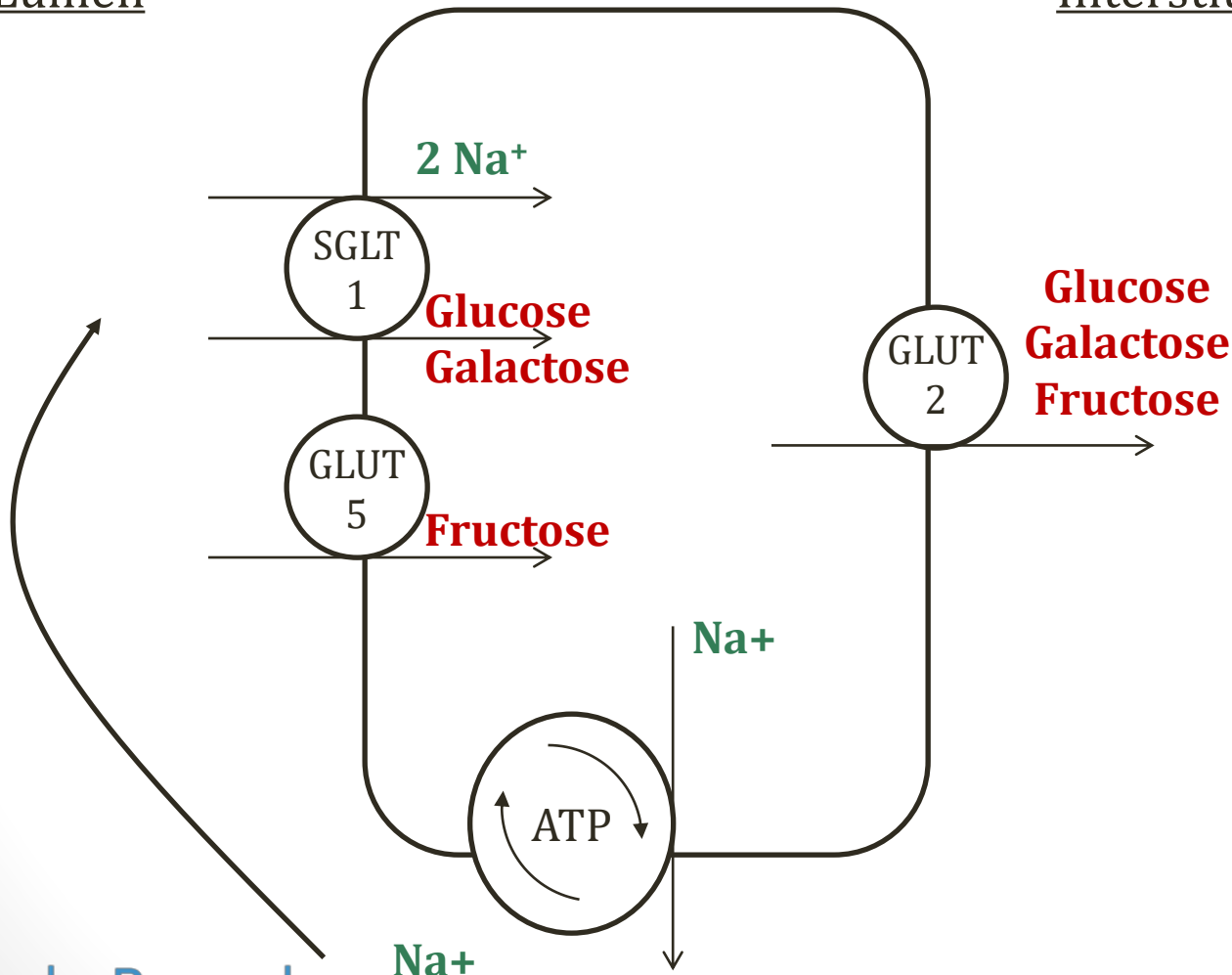
Fructose and Galactose

- Isomers of glucose (same formula: $C_6H_{12}O_6$)
- Galactose (and glucose) taken up by SGLT1
 - Na^+ dependent transporter
- Fructose taken up by facilitated diffusion GLUT-5
- All leave enterocytes by GLUT-2

Carbohydrate GI Absorption

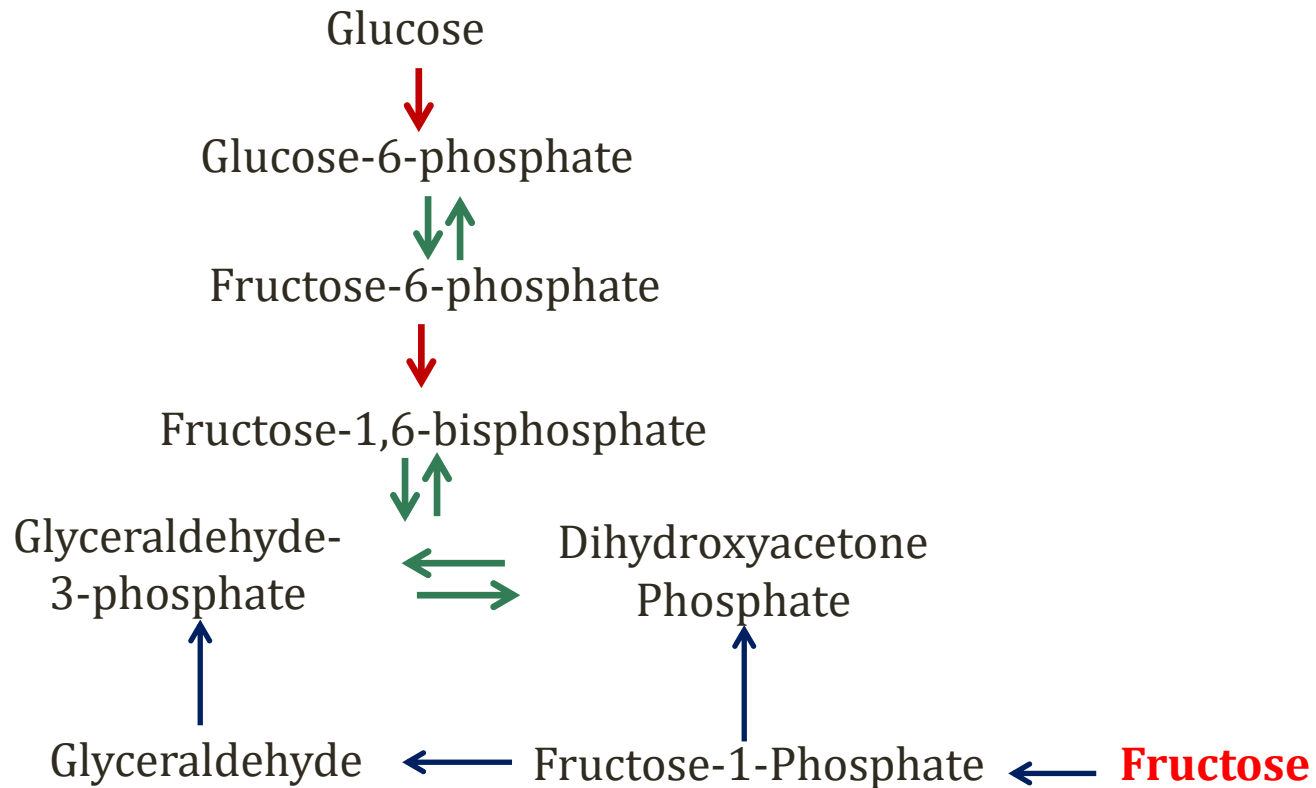
GI Lumen

Interstitium/Blood

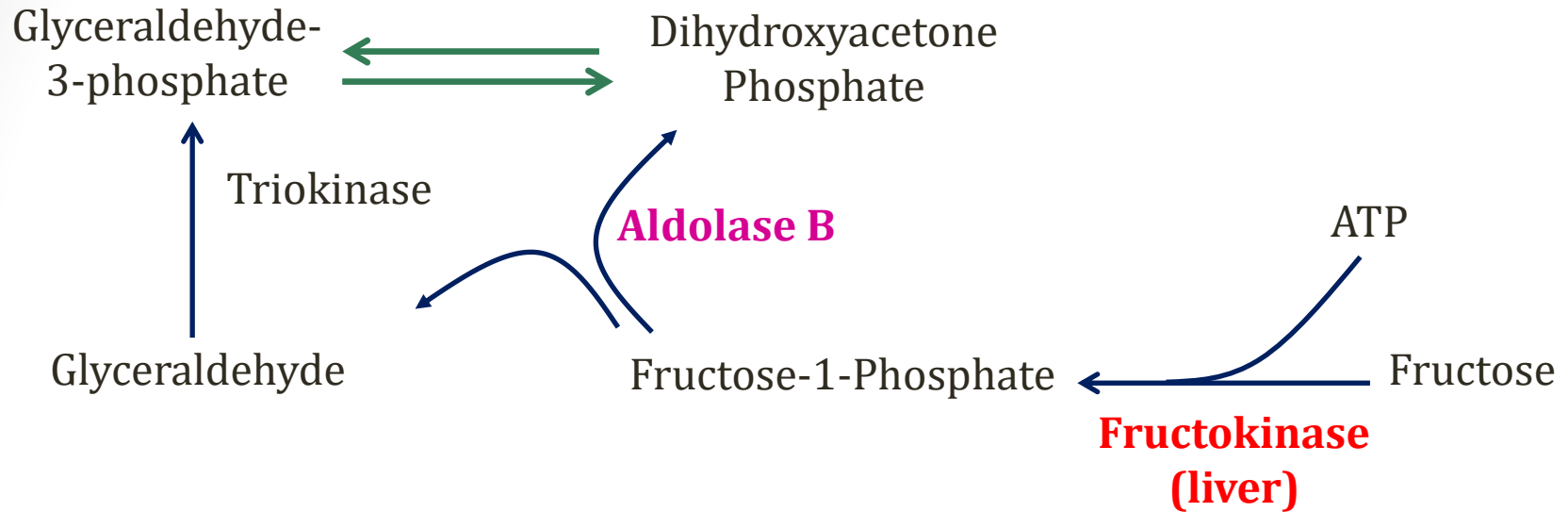


Fructose

- Commonly found in **sucrose** (glucose + fructose)



Fructose



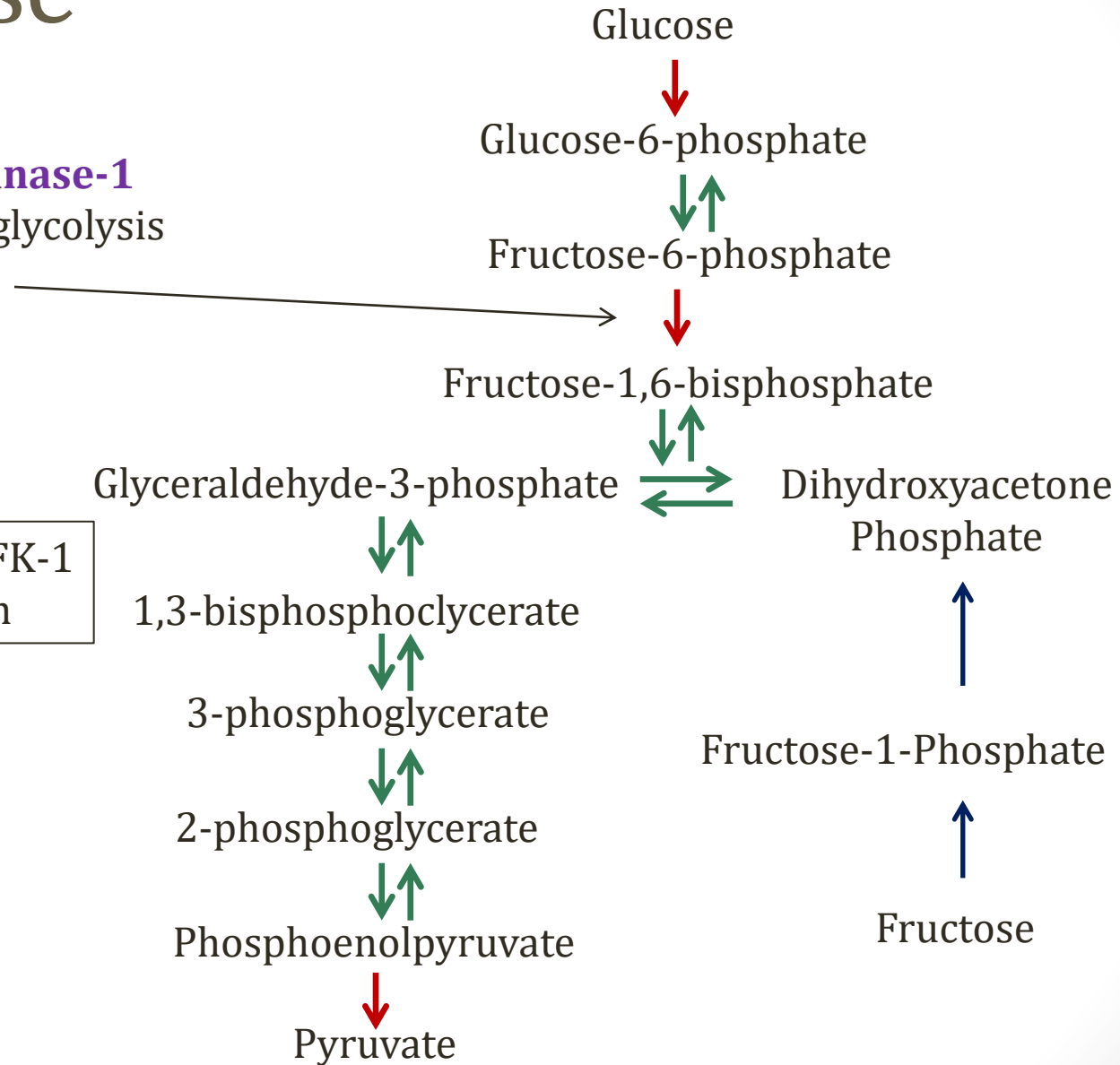
Fructose

Special Point

Phosphofructokinase-1

Rate-limiting step: glycolysis

Fructose bypasses PFK-1
Rapid metabolism



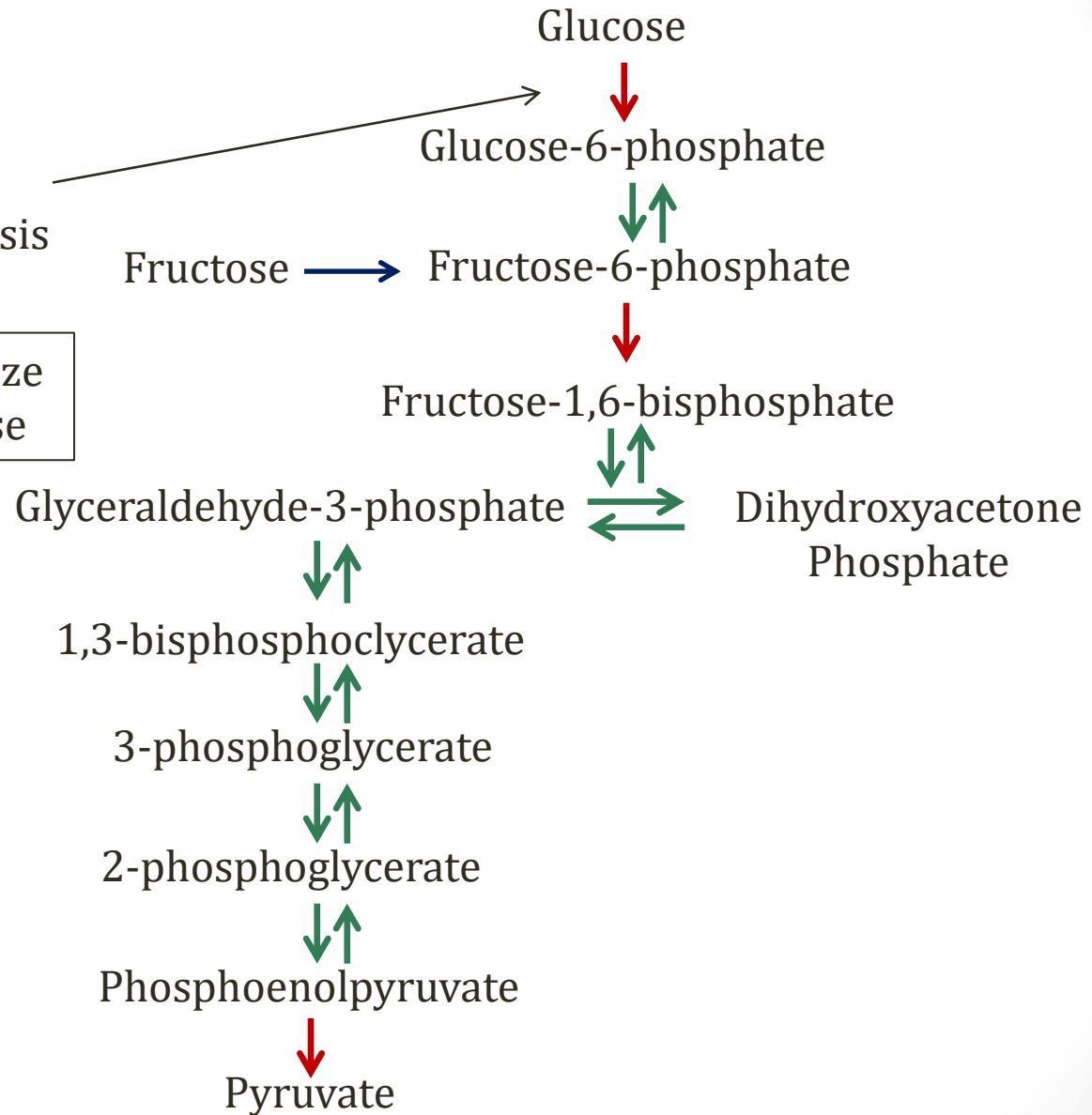
Fructose

Special Point

Hexokinase

Initial enzyme glycolysis

Hexokinase can metabolize small amount of fructose



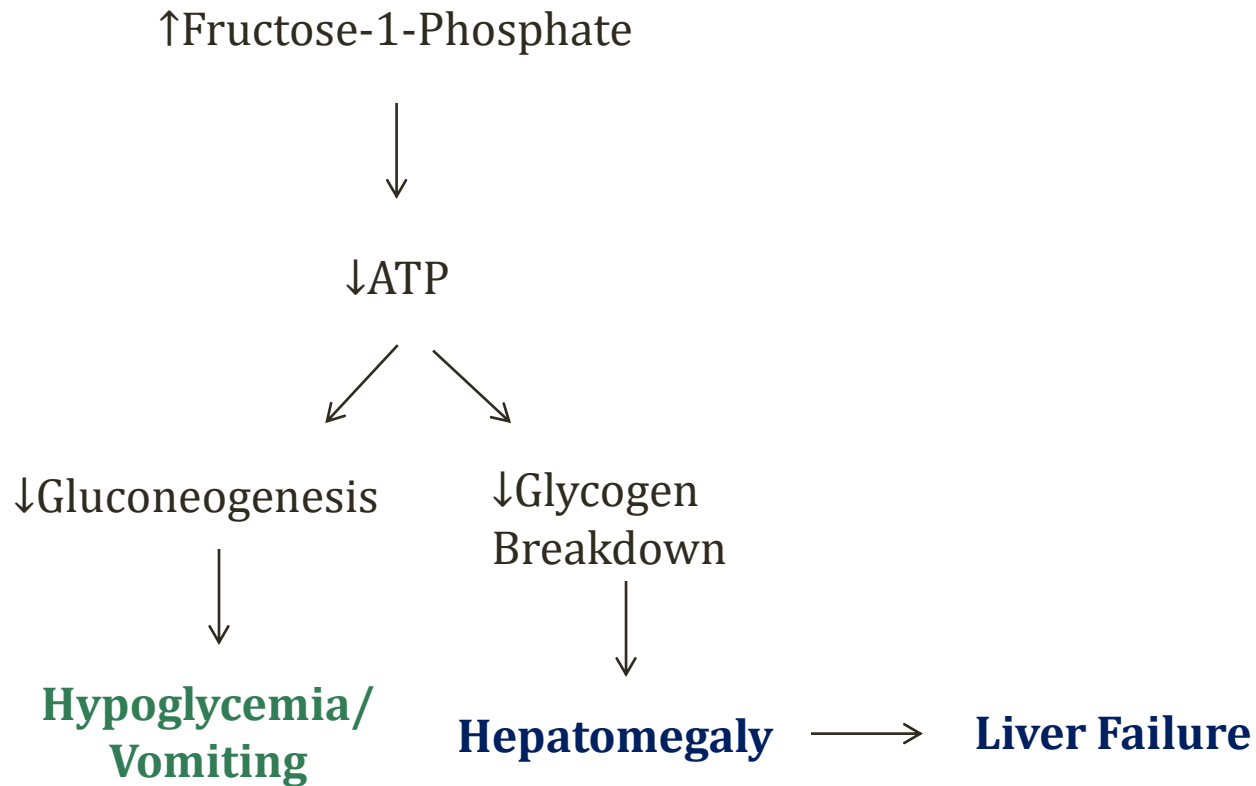
Essential Fructosuria

- Deficiency of **fructokinase**
- Benign condition
- Fructose not taken up by liver cells
- Fructose appears in urine (depending on intake)

Hereditary Fructose Intolerance

- Deficiency of **aldolase B**
- Build-up of fructose 1-phosphate
- Depletion of ATP

Hereditary Fructose Intolerance

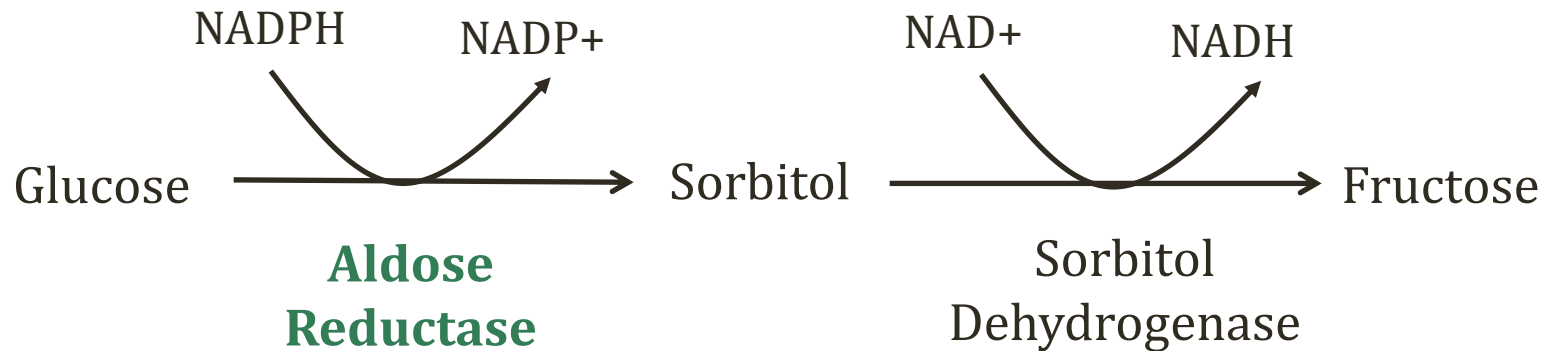


Hereditary Fructose Intolerance

- Baby **just weaned from breast milk**
- Failure to thrive
- Symptoms after feeding
 - **Hypoglycemia (seizures)**
- Enlarged liver
- Part of newborn screening panel
- Treatment:
 - Avoid fructose, sucrose, sorbitol

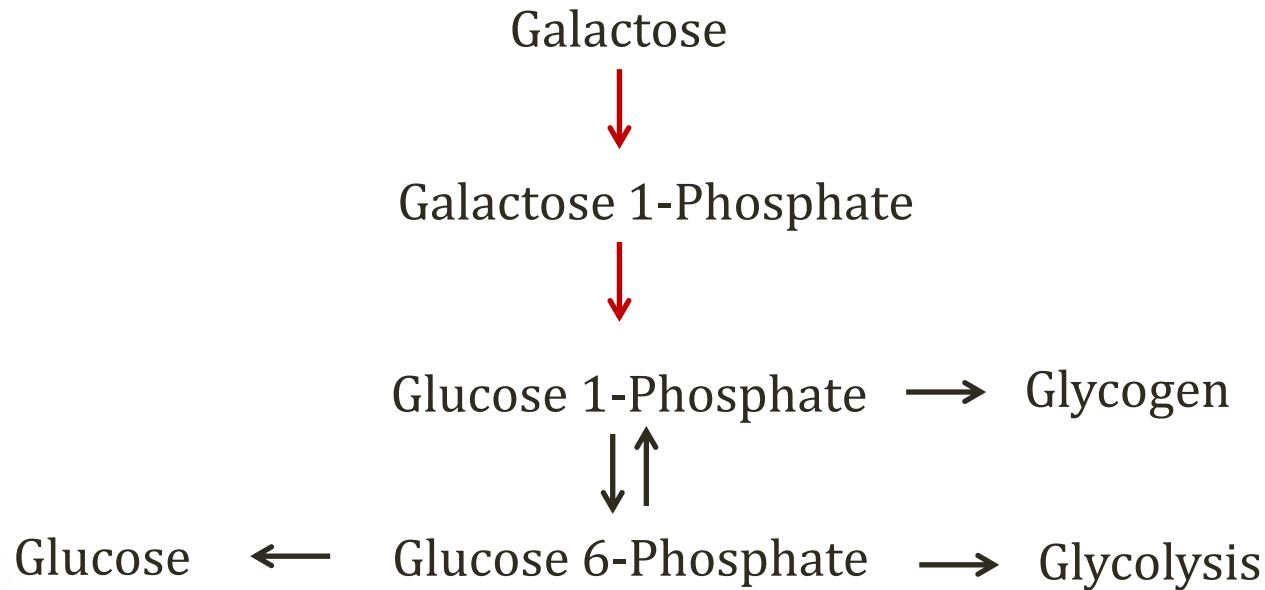
Polyol Pathway

Glucose → Fructose

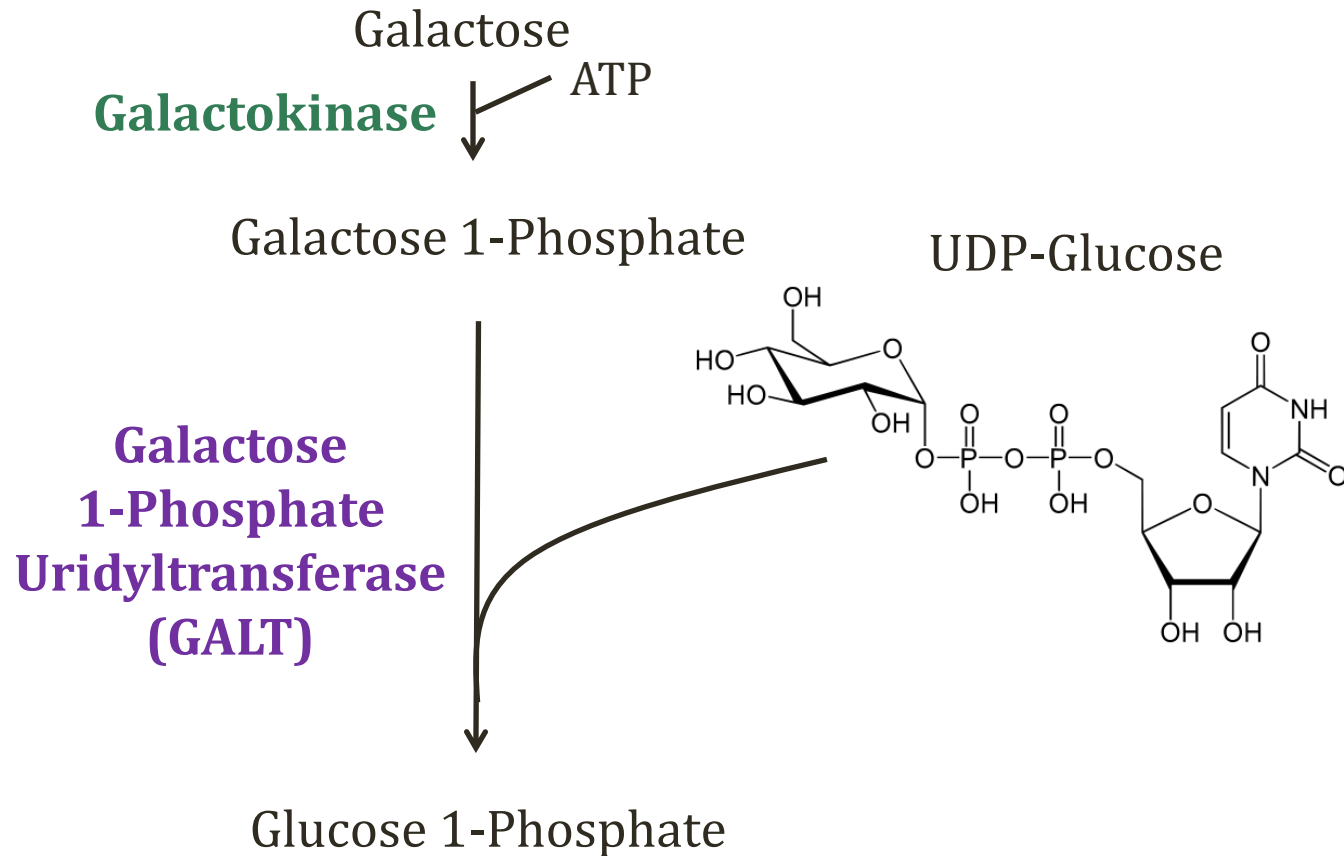


Galactose

- Commonly found in **lactose** (glucose + galactose)
- Converted to glucose 6-phosphate



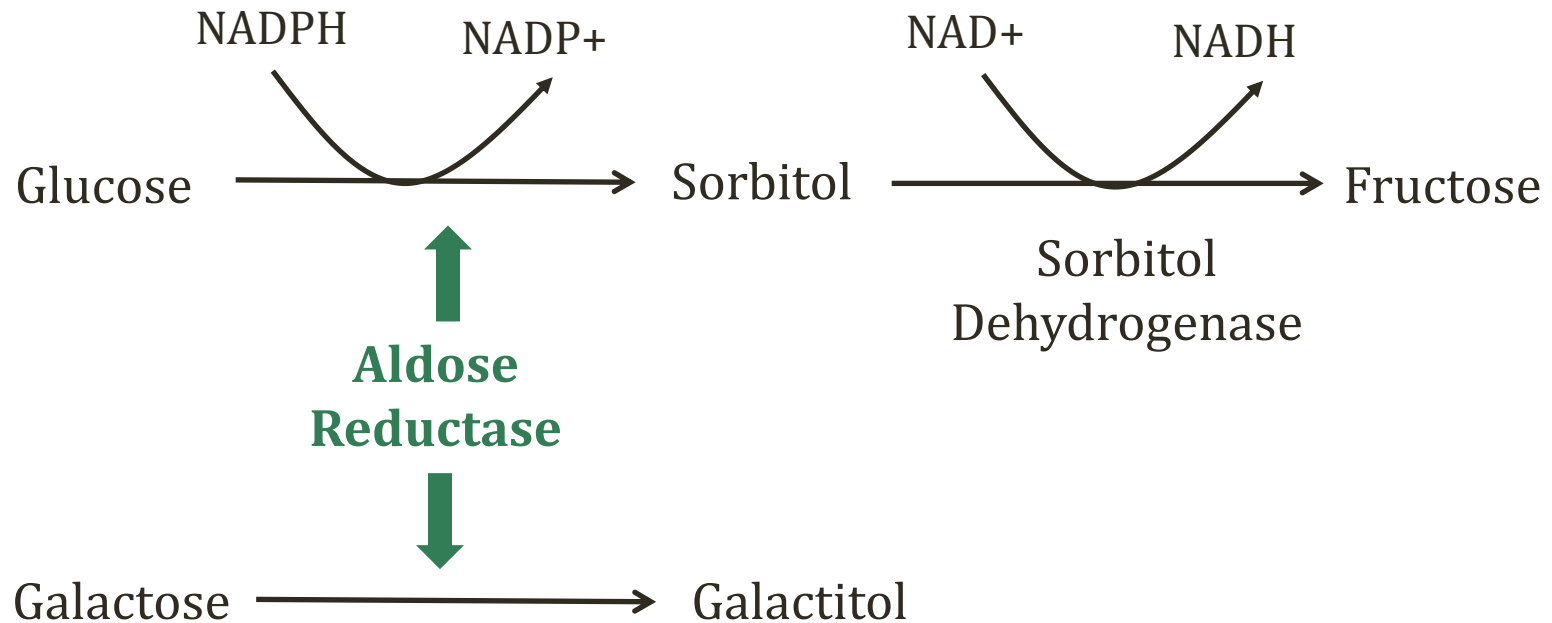
Galactose



Classic Galactosemia

- Deficiency of galactose 1-phosphate uridylyltransferase
- Autosomal recessive disorder
- Galactose-1-phosphate accumulates in cells
- Leads to accumulation of galactitol in cells

Polyol Pathway



Classic Galactosemia



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- Presents in infancy
 - Often first few days of life
 - Shortly after consumption of milk
- Liver accumulation galactose/galactitol
 - Liver failure
 - Jaundice
 - Hepatomegaly
 - Failure to thrive
- Cataracts if untreated



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Classic Galactosemia

- Screening: GALT enzyme activity assay
- Treatment: avoid galactose



Galactokinase Deficiency

- Milder form of galactosemia
- Galactose not taken up by cells
- Accumulates in **blood** and **urine**
- Main problem: **cataracts** as child/young adult
 - May present as vision problems



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